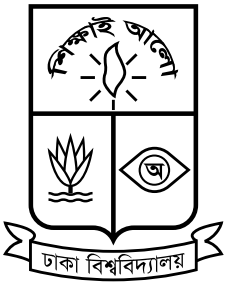
UNIVERSITY OF DHAKA

Department of Computer Science and Engineering



AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

Initial Thesis Progress Report

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Declaration Form

This progress report presents the research foundation, current implementation, and development-stage experimental evidence for the thesis project entitled *AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization*. The work is conducted by the authors under the supervision of Dr. Saifuddin Md. Tareeq. Ideas, methods, datasets, software libraries, and findings originating from prior work are acknowledged and cited. The Drive manifest describes the source package as real SME data, but its provenance has not been independently verified; the report therefore treats results as conditional on the supplied package. The numerical results do not represent a final untouched-test evaluation or a claim of effectiveness across all SME sectors. This report is current to 9 November 2026; the project remains in progress until February 2027.

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Abstract

This progress report documents B-SMART, an AI-powered business analytics and decision-support framework for Bangladeshi SMEs. The Drive manifest claims real SME source data, but it also records a generation seed and "modeled world" validation wording; source provenance has not been independently verified. The complete RQ1 panel contains 561,152 shop–category–day observations and 1,580,361 observed units. In an exposure-conditioned SKU diagnostic, the 28-day exposed-sales mean achieved 183.65% WAPE on 1,130,148 Q4 2025 exposed days, slightly better than the HistGradientBoosting model at 184.66%. Q4 had previously been inspected, so this is not an untouched final test.

For RQ2, source-order reconstruction matched all 72,146 supplied inactivity labels. HistGradientBoosting achieved test PR-AUC 0.746 and ROC-AUC 0.830 on the supplied temporal split. In 2,500 RQ3 scenarios, the trend policy reduced scenario cost by 0.87% but increased scenario lost units from 45 to 100; future oracle-demand values are generated scenario assumptions, not realized sales. A separate eight-cutoff observed-sales backtest is also reported: the trend rule increased weighted assumption-based cost by 0.54% and both policies had zero lost units in the stock-rich sample. A mechanical audit of 1,800 RQ4 cases found complete rule, pack-size and budget consistency, but those cases and their unverified `human_rating` field are not a participant study. The report therefore makes no claim of real-world cost savings, owner-policy benefit or measured usability.

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Chapter 1

Introduction

1.1 Context and Motivation

SMEs generate valuable evidence through sales, purchases, payments, inventory movements and customer transactions, but these records are often distributed across ledgers, spreadsheets and separate applications. Owners may see that sales have changed without knowing which products, customers or operating conditions explain the change, or what response is feasible.

The challenge extends beyond displaying historical KPIs. Retail decisions require demand estimates while accounting for cash, stock, supplier lead time, pack size, storage and expiry. Customer inactivity, incomplete cost records and promotion effects create further analytical risks. This research connects operational events to valid predictions and explainable, constraint-aware decisions [4, 8, 17].

1.2 Bangladesh SME Business Context

The intended setting is Bangladeshi SMEs in retail, wholesale, distribution and related sectors. A shared operational core covers sales, purchases, expenses, payments, customers and performance indicators; inventory, batch, expiry and capacity modules apply where relevant. A common framework does not imply that one model suits every business.

Bangladesh studies identify management support, perceived benefit, cost, complexity and organisational readiness as important influences on SME technology adoption [11, 22]. These findings motivate low-friction data entry, local payment compatibility and owner-readable evidence. Results remain bounded to the businesses and periods represented in the supplied data.

1.3 Decision Environment and Stakeholder Requirements

The owner or manager makes the principal decisions, while operational, finance and supply staff create or validate supporting records. The system must preserve traceability from each result to its source events, transformation rules and historical cutoff. It must distinguish incomplete evidence from a valid zero and allow owners to inspect, modify or reject recommendations.

Table 1.1: Core stakeholder requirements.

Stakeholder	**Responsibility**	**System requirement**
Owner or manager	Reorder, allocate cash and approve customer actions	Feasible options, uncertainty, supporting evidence and owner control
Operational users	Maintain product, sales and stock records	Stable identifiers, validation and correction history
Finance personnel	Reconcile payments, expenses and costs	Traceable transaction links and missing-cost coverage
Supplier contacts	Confirm lead time, pack size and incoming quantity	Dated supply constraints, not assumed universal values
Research and system team	Build and evaluate the artifact	Versioned data, code, cutoffs and reproducible results

The design is assessed on decision relevance, traceability, proportionality and contestability.

1.4 Problem Statement

The central problem is the weak connection between SME operational records, valid analytical estimates and feasible business actions. Four gaps follow:

G1. **Data foundation:** records need consistent identifiers, timestamps, provenance and business boundaries.

G2. **Prediction validity:** features, labels and transformations must respect decision-time availability and chronological evaluation [15].

G3. **Prediction to action:** forecasts and risk scores must be linked to cost, stock, cash, lead-time and consent constraints [1].

G4. **Adoption and accountability:** owners need understandable, traceable evidence and authority to reject or modify recommendations. Feature attribution supports inspection but does not establish causality [16, 18].

The thesis evaluates an integrated SME decision-support framework. It does not assume that machine learning must outperform every baseline; a simpler method is valid when more reliable or economical for the task.

1.5 Analytical Logic and Claim Boundaries

The process separates an observed business event, an estimate based on information available at that time, an eligible action and a later outcome. Missing records weaken the estimate; leakage invalidates it; ignored constraints make an action infeasible; and absent follow-up prevents a credible impact claim.

The thesis distinguishes descriptive KPIs, predictive estimates, prescriptive recommendations and realized business impact. Recorded sales may be censored demand when stock is unavailable. Future inactivity means no observed purchase within a defined window, not permanent customer loss. Model explanations describe model behaviour, not causal effects [16, 18]. Claims are limited to the businesses, periods and tasks actually evaluated.

1.6 Research Aim and Objectives

1.6.1 General Objective

The overall goal is to create and test an AI-based business solution. The analytics framework is designed as a solution that allows Bangladeshi SME businesses to track their business performance, extract reliable and explainable insights, and prioritise feasible strategies while maintaining owner control.

1.6.2 Specific Objectives

The study will:

O1. specify an auditable canonical data model for sales, with a focus on the business and its activities, including products, stock, batches, expiry, purchases, returns, customers, suppliers, payments, and expenses;

O2. build reproducible preprocessing and feature pipelines for performance analysis, SKU demand, customer inactivity, and segmentation;

O3. compare transparent baselines, a pooled multi-business model, and shop-adapted models using chronological and held-out-business validation, with both predictive and decision-oriented measures;

O4. measure uncertainty and provide explanations at both global and case levels, while recognising that association does not imply causation;

O5. formulate a mechanism that retains constraint awareness in reorder, retention, and resource-allocation decisions; and

O6. assess technical correctness, robustness, scalability, usability, and potential business value at distinct levels of evidence, using data collected with consent from the Bangladeshi SME sector.

1.7 Initial Solution Concept

The proposed solution is B-SMART, a multi-sector SME decision-support framework with a shared operational core and configurable analytical tasks. It validates business events, creates time-frozen analytical views, compares baselines with eligible models, and presents estimates with their cutoff, uncertainty and supporting evidence.

Its three functions follow the approved project title: *performance evaluation* reconciles sales, stock and financial indicators; *insight extraction* identifies demand patterns, customer inactivity and operational anomalies; and *strategy optimisation* ranks feasible purchasing or customer actions under business constraints. The owner retains final authority, and outcomes are recorded for audit and monitoring.

1.8 Research Questions

The thesis is based on four research questions:

RQ1.
Do features such as calendar effects, price, promotion, and market context enhance out-of-time branch–SKU demand forecasts over seasonal and statistical baselines in the target business setting?

RQ2.
How accurately and reliably can a historical customer snapshot predict whether there will be no repeat purchase within a predetermined future time period?

RQ3.
Under the same demand and operating constraints, can a forecast-driven reorder policy reduce stock-outs and holding costs compared with the current policy of the participating business?

RQ4.
Do short Bangla action descriptions enhance owner task-friendliness, confidence, and perceived usability without causing information overload or increasing decision time?

1.9 Scope and Boundaries

Empirical claims are restricted to the supplied Bangladesh business dataset and tasks for which its fields are suitable. The current forecasting experiment is at shop–category–day level and predicts observed sales; it does not establish SKU-level latent demand. Customer-risk analysis requires stable pseudonymous identifiers and complete follow-up. Inventory-policy claims require reconciled stock, cost and supplier constraints. Cross-sector generalisation and realized business impact require separate evaluation.

1.10 Expected Contributions and Significance

The evaluated integration of established research methods, constrained within the context and relevance of small SME businesses, is expected to be a research contribution. Specifically, the work is expected to include (i) a traceable business operational-to-analytical data design; (ii) reusable historical-snapshot protocols designed to avoid common forms of data leakage; (iii) a Bangladesh-aware comparison of forecasting and customer-inactivity approaches; and (iv) a clear link between prediction and constrained, actionable recommendations.

The work also provides evidence of the ability of owners to interpret and apply the information presented to them. When making a recommendation, the system presents supporting evidence to help the owner understand the "how and why" of the recommendation, while using familiar rules and maintaining the owner's control over the action. It is not intended to exclude the owner from the decision or to label a numerical output as profitable without considering the business context. Before measuring the impact on the business, the system must first establish that its outputs are accurate, interpretable, and actionable.

1.11 Anticipated Challenges

The main challenges are incomplete fields, stock-ledger anomalies, uneven demand, business heterogeneity, changing conditions and limited independent businesses. These are addressed through eligibility rules, baseline comparisons, temporal validation, uncertainty reporting and proportionate claims. Insufficient evidence is reported as not evaluable, not replaced with assumptions.

1.12 Organization of the Report

Chapter 2 builds up the technical and contextual background, reviews the relevant literature, and identifies the research gap. Chapter 3 introduces the research design, data protocol, formal tasks, proposed framework, and evaluation plan. The Implementation

and Experimental Results chapters document the reproducible pipelines, multi-date validation, model outputs and evidence limits. The Conclusions chapter separates measured findings from claims that still require an untouched holdout, owner-policy replay or participant study.

Chapter 2

Background Study and Related Work

2.1 Background Study

2.1.1 Business Intelligence and the Decision Continuum

Business intelligence (BI) is a blend of data, infrastructure, analytical techniques, best practices, experience, and organisational practice used to enhance decision-making. Chen, Chiang, and Storey trace its evolution from enterprise reporting to web, mobile, and sensor-rich analytics [4]. This thesis follows a three-step approach: *descriptive analytics* determines what happened and where; *predictive analytics* predicts what is likely to happen next; and *prescriptive analytics* compares feasible responses.

The stages are interconnected. An action recommendation is only as believable as the prediction that supports it, and a prediction is only as believable as the operational data from which it was obtained. There are also different integrity requirements for operational and analytical workloads. While a sale requires consistent updates to stock and payment records, an analytical model requires a historical feature view containing values that would not change once the evaluation point has been reached.

The proposed framework therefore separates the transaction record from versioned analytical snapshots. It maintains the connection between them through source identifiers, timestamps, and transformation metadata.

2.1.2 SME Digitalization in Bangladesh

The decision to adopt technology is not only a matter of the accuracy of technology models. A mixed-method study indicates that the important factors determining ICT adoption among SMEs in rural Bangladesh are awareness of the benefits, government support,

top-management support, and financial support [11]. A later survey of 535 SME managers combines the technology–organization–environment model with diffusion-of-innovation concepts. The authors [22] recognize that factors such as relative advantage, complexity, perceived cost, management support, competitive pressure, and government support are important considerations. These studies suggest that organizational readiness and costs, as well as technological fit and workflow functionality, warrant attention when considering technology adoption. However, the two studies do not evaluate a retail prediction system.

This operating context also includes mobile financial services. Through in-depth interviews, accessibility, spatial and temporal mobility, and access to information were found to be important issues in the daily use of MFS in Bangladesh. Self-control and the affordances related to self-control are also important considerations [9]. Therefore, the proposed data model includes the payment mode in the settlement status field, along with other common business fields. The study should not be construed as providing evidence of better business performance for a named provider or of better performance of one local provider compared with another.

2.1.3 Shared SME Core and Sector-Specific Requirements

The research scope covers SME businesses across sectors. Shared entities support comparable performance indicators, while sector-specific extensions capture differences in products, service capacity, sales cycles, procurement and operating constraints. Inventory-based firms may require SKU, stock, pack, batch and shelf-life fields; service firms may require service type, bookings, staff availability and delivery capacity. Fields that do not apply are marked not applicable rather than populated with invented values.

The artifact supports business performance evaluation, insight extraction and feasible strategy recommendations. Owners retain authority over actions. Sector-specific requirements and owner constraints are assessed before an analytical task is enabled; unsupported tasks return an explicit evidence gap.

2.1.4 Performance Evaluation and Data Foundations

Well-defined measures are the starting point for performance evaluation. Revenue, gross margin, average order value, stock turnover, return rate, and customer repeat rate each need to include a defined time window, an aggregation method, and appropriate treatment of events such as rescheduling, refunds, and lost fees. If any of these elements are not incorporated into a dashboard value, the resulting measure may be misleading or ignored. Definitions may also differ from the accounting records or may refer to different time periods. The canonical layer therefore maintains invoice lines and stock movements while creating analytical measures through documented views.

Data provenance concerns where a record came from, when it was received, and how it was transformed. Primary business data should be accompanied by provenance information,

together with appropriate customer anonymisation and organisational anonymisation. For public data, the context and reasonable implications of the applicable licence should also be documented.

2.1.5 Data Quality, Provenance and Temporal Semantics

Analytical validity depends on the meaning and timing of each field. Event time, recording time, correction time and decision cutoff are not interchangeable. Historical features must use only information available by the cutoff; later corrections or outcomes must not leak into earlier snapshots [15]. Missing records are not automatically zero, and unknown stock exposure remains unknown.

A reproducible manifest records source, owner, collection period, schema, checksum, row counts, exclusions and transformation rules. Reconciliation checks keys, duplicates, quantities and links among orders, sales lines, payments and inventory movements. These checks establish internal consistency; source documentation establishes declared origin.

2.1.6 Customer Analytics: RFM, CLV and Segmentation

A simple summary of *recency*, *frequency*, and *monetary value* is provided by RFM. It summarises a customer's purchase or transaction history using features that are easy to calculate and maintain an unambiguous behavioural interpretation. Fader, Hardie, and Lee link RFM with Customer Lifetime Value (CLV) through the Pareto/NBD and gamma–gamma components, demonstrating how the value of a particular customer history can be estimated [7]. CLV can be used as an outcome or planning measure, but it is not necessarily suitable as a direct prediction target or operational goal.

A field that defines a segment should not be admitted as an independent predictor of that same segment, as this would introduce information leakage. The question for unsupervised segmentation is whether coherent behavioural groups can be identified without an existing label. K-Means aims to:

$$\underset{\{C_k,\mu_k\}_{k=1}^{K}}{\text{minimize}} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2. \tag{2.1}$$

The procedure returns clusters even when the data form a continuum. Therefore, both the number and the interpretation of the clusters need to be validated. The silhouette coefficient measures the similarity of observations within a cluster and their separation from the closest other cluster [20]. The stability of the clustering parameters across resampling and the business usefulness of the resulting clusters should be treated as separate tests. An attractive label such as "loyal" does not, by itself, imply statistical stability or actionability of a cluster.

2.1.7 Retail Demand Forecasting

A demand forecast estimates the number of units needed for a specific product, location, and future horizon. Retail demand series can be influenced by price, promotion, stock-outs, festivals, assortment changes, and aggregation across products or branches. Fildes, Ma, and Kolassa emphasize that retail forecasting practice cannot be simplified into a single algorithm because of data hierarchy, operational requirements, and the amount of data involved. The forecasting process and the way forecasts are used also matter, alongside statistical accuracy [8].

Intermittent SKU demand contains numerous zero-demand periods between non-zero observations. Croston's method smooths the non-zero demand size and the interval between demands separately. Syntetos and Boylan examine the accuracy and bias of intermittent-demand estimates [24]. This work supports the use of an intermittent-demand baseline rather than requiring a uniform and constant baseline. For dense time series, an appropriate dense-series model can be used at the SKU level. Forecasts also need to be reconciled across different levels of aggregation, including product, category, branch, and overall business totals. Optimal-combination reconciliation provides a principled means of generating coherent hierarchical forecasts [12].

When modelling, the model should be only as complex as the available evidence supports. ARIMA-family models represent serial dependence, including dependence after appropriate transformations; tree ensembles can learn nonlinear effects from lagged, calendar, and price variables; and sequence models can use ordered historical observations. Random Forest and XGBoost provide suitable tabular alternatives [2, 5]. The M5 retail competition illustrates the importance of large-scale, hierarchical, and uncertainty-aware comparison [17]. A seasonal-naive forecast is therefore a necessary benchmark rather than a ceremonial baseline.

Although inventory usefulness is related to forecast accuracy, the two are not the same. Hyndman and Koehler discuss the limitations of common percentage-error measures and suggest alternative approaches, including MASE as a scale-free measure for comparison [13]. A study by Syntetos and colleagues associates the choice of forecasting methods with stock-control requirements, whereas forecasting selection based solely on statistical accuracy may not reflect the needs of inventory management [23]. The methods are therefore evaluated using point-forecast error, uncertainty, and simulated stock-out and holding costs rather than selecting a technique based on a single easily computed metric.

2.1.8 Forecastability, Demand Regimes and Decision Loss

Retail series differ in volume, variability, intermittency, trend and seasonality. ADI and squared coefficient of variation describe regimes, but labels must use only history available at the forecast origin and should diagnose rather than automatically route models. Croston-family methods and TSB are relevant intermittent-demand comparators [6, 24, 27].

Forecast accuracy and inventory performance are distinct. Lower WAPE may not reduce shortages or holding cost when bias, lead time, pack multiples, cash and stock constraints are ignored. Models should be compared on common origins and targets, then evaluated through a separate policy replay [23].

2.1.9 Customer Inactivity, Class Imbalance and Calibration

Churn can be a documented cancellation for a subscription-based service. For non-contractual retail, however, churn has to be specified at a cutoff time t as no repeat purchase within a given time interval $[t, t + H]$. All features must be known at time t, and the training and validation processes must not be contaminated by information from the future label window. This definition translates a business term into a repeatable prediction task.

In business-to-business settings, the positive class can be scarce or vary across datasets. Saito and Rehmsmeier provide a rationale for the superiority of precision–recall analysis over ROC analysis for imbalanced binary classification tasks [21]. Model comparison should therefore report class prevalence and the quality of the predictions before considering action costs and benefits.

The prescriptive layer includes an additional requirement. When using a model output in a decision calculation, ranking alone is not sufficient; the output needs to be converted into an interpretable probability or score. A probability of 0.7 should, under appropriate calibration, correspond approximately to a 70% frequency of the event among comparable cases. Niculescu-Mizil and Caruana examine the calibration of probability estimates from supervised learners and post-hoc calibration methods [19]. The proposed framework therefore reports Brier score in addition to PR-AUC, ROC-AUC, and top-fraction lift, while also making calibration information available.

2.1.10 From Customer Score to Responsible Retention Decision

A customer-risk model does not grant permission for an action; instead, it provides a ranking or probability. For a given contact capacity, ranking quality determines whether the selected fraction contains more future inactive customers than an untargeted policy. When an expected value is calculated, the probability needs to be calibrated because the benefits and costs are subject to uncertainty and are weighted by the estimated probability. Therefore, the operational threshold should be determined using validation data and the pre-declared assumptions. The final test set should not be used to determine contact capacity or utility assumptions; rather, it should be used to confirm the choices made during the development process.

Additional ambiguity is created by the customer record. The frequency and amount of purchases may be linked to opportunities to visit, household purchasing patterns, or shared identifiers. Neither return behaviour nor discount-seeking behaviour alone is sufficient to

indicate loyalty or future value. Fader, Hardie, and Lee demonstrate why combined transaction history can provide more information than a single RFM label [7]. Segmentation can support interpretation, but a persuasive "segment name" is only a description of a behavioural pattern. It does not establish that the pattern is statistically stable. Therefore, clustering should be accompanied by stability checks and owner review before the resulting segment output is used for decision-making [20].

An independent feasibility gate is also required for responsible use. A score indicating a high probability of inactivity does not, by itself, justify customer contact if consent, contact policy, or an appropriate offer is absent. The score should also never be interpreted as a causal attribution, such as "long recency increased risk," or restated as a causal claim that a particular message will retain the customer. The thesis therefore treats the prediction of inactivity, the reliability of the estimated probability, contact eligibility, and subsequent customer response as separate links in the decision process. This structure allows the model to be scientifically informative even when the available evidence is insufficient to support a claim about profit or the causal effect of a retention action.

2.1.11 Leakage, Explainability and Change Over Time

Data leakage occurs when training is exposed to information that would not legitimately exist at prediction time. It can enter through target-derived features, customers who appear on both sides of a data split, post-customer-action information, post-outcome variables, or transformations that are fitted before the data are split [15]. Leakage control is therefore a characteristic of the data pipeline, not a model option. Before training, the system should define the prediction time, feature-availability rules, and data-splitting policy.

SHAP provides additive feature contributions relative to a reference value and supports both local and global inspection of model predictions [16]. Such an attribution describes how the fitted model uses encoded features; it does not establish a causal relationship. In social-science research, as synthesized by Miller, the focus is often on useful explanations that are selective and contrastive [18]. Accordingly, a recommendation card should respond to the question, "Why this action now?" by presenting a small number of salient, decision-relevant factors. It should avoid causal language while providing sufficient supporting evidence for the owner to assess the recommendation.

Business conditions also evolve over time. Retail monitoring therefore needs to distinguish between data-quality failure, input-distribution shift, and deterioration in outcome performance. A model should not be considered valid simply because it remains on a retraining schedule. Evidence that a model remains valid should come from continued evaluation under a declared validation protocol. Any retrained model should be evaluated using the same validation principles that were established during development, with changes explicitly documented.

Once a prediction is made, the next step is to take an action subject to business constraints. Prescriptive analytics incorporates predictions into an explicit decision problem. Bertsimas and Kallus develop prescriptions based on data with auxiliary information, where observations are used to inform conditional stochastic optimization [1]. The present thesis follows the underlying principle of evaluating actions according to expected benefit, cost, and constraints. It does not claim to reproduce the asymptotic method or theoretical approach of the cited work. Instead, reorder and retention decisions are treated as separate action families because their outcomes, costs, permissions, and uncertainties differ.

The proposed system is both an information-system artifact and an analytical study. Design-science research develops and evaluates artifacts in relation to a relevant problem and an existing knowledge base [10]. This perspective contributes to an evaluation plan that considers predictive validity, software behaviour, usability, and the consequences of decisions made by the owner, rather than treating a single model metric as the definitive criterion of success.

2.1.12 Cross-Stream Evidence Synthesis

The literature supports five connected requirements: reliable operational data, time-valid prediction, appropriate baselines, constraint-aware action and accountable owner use. Bangladesh adoption studies inform local design; forecasting and customer-analytics studies inform model evaluation; leakage, calibration and explanation research inform validity; and prescriptive analytics links estimates to decisions. The contribution is their integration and evaluation within the available SME evidence, not a newly invented algorithm.

2.2 Literature-Review Procedure

This is a focused background review, not an exhaustive systematic review. Sources were selected for relevance to Bangladesh SME digitalisation, business analytics, customer behaviour, retail forecasting, inventory, leakage, calibration, explainability and prescriptive decisions. Peer-reviewed studies, established methods and identifiable dataset repositories were prioritised. These sources provide contextual and methodological foundations; they are not substitutes for the empirical business records analysed here.

2.3 Research Gap and Thesis Positioning

Six connected gaps motivate the study:

G1 – Contextual evidence.
Bangladesh SME adoption research explains readiness and barriers, while large retail studies provide forecasting evidence; neither alone establishes transaction-level effectiveness in the target operating context.

G2 – Operational integration.
The thesis preserves the path from source event and correction history to frozen snapshot, prediction, action and outcome.

G3 – Joint methodological validity.
Leakage control, intermittent demand, hierarchy, forecast metrics and calibration must be applied together.

G4 – Prediction to decision.
Forecasts and customer scores must be translated into feasible actions under cash, stock, lead-time, cost and consent constraints.

G5 – Owner-centred explanation.
Feature attribution alone does not show that an owner understands or appropriately uses a recommendation.

G6 – Lifecycle accountability.
Model versions, recommendations, owner responses, outcomes and revalidation must remain linked over time.

The central gap is integrative and evaluative: whether an auditable, multi-business SME system can combine temporally valid analytics with feasible, understandable actions while retaining evidence of later outcomes. The thesis tests this bounded contribution against the supplied data and available businesses.

Table 2.1: Research gaps, responses and required evidence.

Gap	Response	Evidence required
G1 Context	Use supplied Bangladesh SME records	Manifest, source reconciliation and bounded results
G2 Integration	Versioned events and historical snapshots	Traceability from records to features, outputs and actions
G3 Validity	Chronological splits, common baselines and frozen evaluation	Out-of-time metrics, ablations, calibration and leakage checks
G4 Decision	Constraint-aware purchasing and retention logic	Policy replay using reconciled costs and constraints
G5 Explanation	Owner-readable evidence, uncertainty and source details	User-task correctness and appropriate interpretation
G6 Lifecycle	Versioning, monitoring and outcome tracking	Revalidation and recommendation-outcome records

The contribution is a reproducible process for selecting suitable methods, linking estimates to operational constraints and limiting claims to the evidence actually tested.

Chapter 3

Proposed Methodology

3.1 Multi-Sector Applicability and Validation

The intended scope is SME business analytics across sectors, rather than one industry. A common core models organizations, branches, transactions, expenses, payments, customers and suppliers. Configurable sector extensions preserve differences in products, services, inventory, resource capacity and operating rules. Demand and reorder experiments apply to inventory-based firms; service firms require service-demand and capacity tasks. Customer-risk tasks require stable pseudonymous identifiers and adequate follow-up in either setting.

Sampling and reporting will retain sector and business identifiers. Results will be reported per business and per sector before aggregation. Pooled models will be compared with sector-specific and business-adapted alternatives using the same chronological cutoffs. Held-out-business tests assess transfer within represented sectors; held-out-sector tests assess transfer across sectors when enough independent sectors and businesses are available. In their absence, cross-sector transfer remains untested. Preprocessing is fitted only on training data, and firms with many rows must not dominate business-level conclusions. No universal accuracy or business-value claim follows from framework coverage.

3.2 Research Approach

3.2.1 Research Strategy and Design-Science Rationale

This thesis adopts a design-science research approach supported by empirical evaluation. Design science is appropriate because the study develops an artifact intended to address an applied organizational problem and evaluates that artifact against explicitly defined requirements [10]. The principal artifact is the B-SMART framework, including its canonical operational data design, time-aware analytical pipeline, constraint-aware recommendation process, and mechanisms for presenting supporting evidence to the owner.

Empirical evaluation is necessary because a plausible technical architecture alone is insufficient to establish predictive validity, business usefulness, operational feasibility, or usability. Accordingly, the research evaluates not only whether the proposed system can be implemented, but also whether its analytical outputs remain temporally valid, whether its recommendations respect the operating constraints of participating businesses, and whether intended users can interpret and appropriately use those recommendations.

3.2.2 Research Stages and Units of Analysis

The work follows four stages: define the problem and evidence; design the operational schema, snapshots, models and decision rules; evaluate data integrity, prediction and robustness; and assess decision consequences and owner use. The four stages are conducted sequentially, with feedback permitted during development but not after the final test is frozen. Later findings may refine the artifact but cannot alter a frozen final test.

The unit of analysis depends on the task: a shop–category or SKU time series for forecasting, a pseudonymous customer snapshot for inactivity, and an eligible action under declared business constraints for strategy evaluation. Evidence from one unit is not treated as proof at another level.

3.2.3 Design Requirements and Acceptance Logic

The artifact is assessed against eight requirements:

DR1. **Decision relevance:** identify user, unit, horizon and decision.

DR2. **Operational integrity:** preserve identifiers, event times and corrections.

DR3. **Temporal validity:** use only information available at the cutoff.

DR4. **Proportionate modelling:** compare every candidate with a baseline.

DR5. **Uncertainty:** report intervals, calibration and sensitivity where relevant.

DR6. **Feasible action:** enforce stock, cash, lead-time, pack, expiry and consent constraints.

DR7. **Human authority:** allow owners to inspect, modify or reject actions.

DR8. **Reproducibility:** retain data, feature, model and output versions.

Acceptance is non-compensatory: accuracy cannot compensate for leakage, privacy failure or infeasible action. Requirements are classified as met, partially met, not met or not evaluable. Missing evidence reduces the strength of a conclusion rather than being replaced by an assumption.

3.2.4 Research Questions, Hypotheses, and Measures

The research questions introduced in Chapter 1 are operationalized as follows:

RQ1 / H1

A business demand model incorporating locally available calendar, price, promotion, and market-context features will reduce out-of-time WAPE or MASE relative to both an otherwise identical model without those feature groups and a seasonal-naive benchmark.

RQ2 / H2

A model trained from historically valid customer snapshots will predict future inactivity better than an event base-rate or rule-based baseline, as measured by PR-AUC, and will achieve Lift@10% greater than 1.0 without unacceptable deterioration in probability calibration.

RQ3 / H3

Under identical demand paths, costs, and operational constraints, the proposed forecast-based replenishment policy will reduce the combined consequences of stockouts and inventory holding relative to the current replenishment policy used by the participating owner.

RQ4 / H4

In counterbalanced owner decision tasks, an explanation-supported condition will improve decision accuracy or confidence without causing a practically detrimental increase in task-completion time relative to an output-only condition.

Table 3.1: Alignment of research questions, evidence, comparisons, and primary measures.

RQ	Unit and Comparison	Primary Measures	Evidence Required
RQ1	Unseen branch–SKU periods; context-aware model versus ablated models and forecasting baselines	WAPE, MASE, interval coverage, and related out-of-time forecast measures	Time-stamped sales, stock, price, promotion, calendar, and relevant context data
RQ2	Future customer outcome windows; predictive model versus base-rate or rule-based baseline	PR-AUC, Lift@10%, Brier score, and calibration measures	Pseudonymous customer transactions with historically valid feature cutoffs and sufficient future follow-up
RQ3	Identical demand paths; proposed replenishment policy versus the participating owner's existing policy	Stock-out units, inventory holding, and total cost under declared assumptions	Stock records, lead times, purchasing constraints, and validated cost assumptions
RQ4	Equivalent decision situations; explanation-supported output versus output-only condition	Decision correctness, completion time, confidence, and perceived usability	Consenting target users and counterbalanced decision tasks

The final test set will not be inspected during hypothesis development, model selection, feature selection, or primary metric selection. Measures introduced after this stage will be identified as exploratory. Furthermore, a statistically detectable difference will not automatically be interpreted as practically important unless its magnitude, operational consequence, or cost implication is material to the target decision.

3.2.5 Target Population and Sampling Plan

The target population consists of SMEs operating in Bangladesh across retail, wholesale, distribution and service sectors. Recruitment will seek multiple independent businesses and record sector, size and operating model. An initial three-to-five-business pilot can test collection and usability, but cannot establish population-wide or cross-sector effectiveness. Broader sector comparisons require adequate independent businesses in each stratum.

The intended analytical target is approximately 12–24 months of longitudinal records across participating businesses, subject to data availability, data quality, owner consent, and ethics approval. A business with substantially less history may still contribute to testing data collection, mapping, schema, and data-quality procedures, but such a business shall not independently determine claims of temporal or cross-business generalization.

Businesses will be purposively recruited because participation depends on access to relevant longitudinal evidence, including invoice, stock, purchase, batch, and expiry information where available, together with owner agreement. At a minimum, included transaction

data shall contain interpretable dates, product identifiers, quantities, and sufficient contextual information to reconstruct the relevant business process. Participating businesses shall also have an owner or responsible operator capable of describing the current replenishment process.

Customer-level analysis imposes additional eligibility requirements. It requires stable pseudonymous customer identifiers and sufficient follow-up after each historical cutoff to determine the relevant outcome. If a business, analytical task, or time period does not satisfy the minimum evidence requirements for a specific analysis, that unit will be excluded from that analysis rather than being repaired through invented identifiers, fabricated constraints, or assumed outcomes.

3.2.6 Data Sources and Evidence Tiers

Evidence will be separated into explicit tiers so that convenient external, public, or generated data cannot silently alter the claim being tested.

Table 3.2: Planned evidence tiers and corresponding limits on inference.

Tier	**Source**	**Authorized Use**	**Prohibited or Unsupported Inference**
1	Consenting Bangladesh business invoice, stock, purchase, batch, expiry, customer, and constraint records	Primary business forecasting, customer-risk, strategy, and usability evaluation	Generalization beyond sampled businesses or untested transfer across SME verticals
2a	Bangladesh market-price data, including relevant WFP/DAM sources [25]	Local external context and separate market-price analysis	Direct inference of SKU-level business sales, customer behaviour, or business revenue when those variables are not observed
2b	UCI Online Retail II data from the United Kingdom [3]	Real-invoice pipeline testing and portability checks for selected methods	Claims about Bangladeshi purchasing behaviour or local business effectiveness
2c	Public Bangladesh retailer demand-series data [14]	Local quantity-forecast procedure and implementation checks	Multi-SKU, customer, price, inventory, or profit claims when the required fields are absent
3	Software test records	Software testing, rare-case testing, scalability checks, and privacy-safe examples	Claims of field effectiveness, real customer behaviour, or local operational impact

Each incoming dataset will be accompanied by a manifest recording its source, owner or repository, collection period, receipt date, checksum, licence or agreement, row count, schema version, mapping rules, exclusions, and known limitations.

Tier-3 generated data will never be mixed into a Tier-1 test set. If generated or augmented records are used during model development, results from real-only training and augmented training will be reported separately and evaluated against the same untouched real test period wherever the required real evidence is available.

3.2.7 Data Acquisition and Variable Availability

The supplied package contains tables for sales, customers, products, purchases, payments, inventory, returns, promotions and calendar records. Its manifest claims real-business origin, but also includes a generation seed and a "modeled world" validation description. The provenance of the underlying transactions and people has not been independently established. RQ1 uses the complete daily shop–category observed-sales panel; other tables are joined only after their keys, event times and business meaning are checked.

Each variable is classified by availability time. Price, promotion or supplier conditions are predictors only when known at the forecast origin. Later sales, returns and inventory movements are outcomes, not predictors. Raw business files remain in controlled storage; the repository contains code, schemas, checksums and aggregate results.

3.2.8 Data-Quality Assessment and Task Eligibility

The ingestion audit validates required fields, types, identifiers, duplicate keys, date ranges, non-negative quantities and cross-table relationships. Sales lines are reconciled to parent orders; inventory sale movements are checked against sales lines; and daily aggregates are compared with transaction-level reconstruction. Mismatches are reported and quarantined, not silently repaired.

For the complete shop–category panel, the audit checks unique shop–category–date keys, required values, consistency of the zero-sales flag and a complete daily calendar. This panel is eligible for observed-sales forecasting at its declared aggregation level. It does not automatically make sparse SKU exports eligible for zero-demand labels.

SKU exposure requires evidence that the product was active, the shop was operating and stock was available. Missing SKU dates are not automatically zero. Negative reconstructed balances, incomplete inventory coverage, unknown assortment, closed periods and stock-out periods remain censored or unknown until reconciled. RQ1 observed-sales forecasting can proceed on the complete panel; SKU latent-demand estimation, customer inactivity and inventory-policy replay each require their own validated inputs.

3.2.9 Ethics, Privacy, and Governance

University ethics approval and an appropriate signed data-sharing agreement will be obtained before primary data collection begins. Direct identifiers such as customer names, phone numbers, email addresses, national identifiers, account numbers, or mobile-wallet numbers are not required for the research analytical tables.

Where customer-level analysis is permitted, each participating business will use a separate salted pseudonymous customer key. Any mapping capable of re-identification will remain under the control of the participating business and will not be stored in the research repository.

The data-sharing agreement will specify the research purpose, authorized researchers, storage arrangements, retention period, publication format, withdrawal conditions, and deletion procedures. Access to research data will follow least-privilege principles, while published outputs will consist of aggregated results or individually reviewed de-identified examples.

A model-generated customer-retention recommendation does not itself authorize contact with that customer. Customer consent and the participating business's contact policy remain independent feasibility constraints. Similarly, system outputs are intended to support rather than replace the decision-making authority of the owner.

3.2.10 Canonical Operational and Analytical Model

The operational model links businesses and shops to products, customers, suppliers, sales orders, sales lines, payments, purchase orders, inventory movements, returns and promotions. Stable identifiers preserve relationships. Analytical tables are derived at a declared grain and cutoff; they are not independent source-of-truth records.

The implemented RQ1 grain is shop–category–day. Its target is observed units sold. SKU-level demand requires separate assortment and stock-exposure validation before missing dates can be interpreted.

3.3 Formal Problem Formulation

Let b denote a business, l a branch, p a product, r an inventory batch, c a customer, and t a decision time. Let

$$D_t = \left\{ \begin{array}{c} \text{Sales, Stock, Batches, Expiry, Purchases, Expenses,} \\ \text{Customers, Constraints, and Context} \end{array} \right\}_{\leq t}, \tag{3.1}$$

denote the governed evidence available at or before decision time t. Accordingly, any analytical output produced at t must be derived only from information contained in D_t.

3.3.1 Performance Evaluation

For each KPI K, the system returns $K(D_t; W, g)$, where W is the reporting window and g is the aggregation level. Each KPI retains its numerator, denominator, filters, definition version, and data-quality status so that the owner can reproduce and inspect the calculation.

For reporting window W, net sales are calculated from valid invoice lines using the documented business treatment of discounts, cancellations, and returns. A general form is

$$\text{NetSales}_W = \sum_{i \in W} q_i p_i - \sum_{i \in W} d_i - \sum_{r \in W} v_r, \tag{3.2}$$

where $q_i p_i$ is the gross value of invoice line i, d_i is the recognized discount, and v_r is the value of a valid return or reversal attributed to the reporting window.

Gross margin is reported only when sufficiently reliable cost evidence is available:

$$\text{GrossMarginRate}_W = \frac{\text{NetSales}_W - \text{COGS}_W}{\text{NetSales}_W}, \tag{3.3}$$

provided that $\text{NetSales}_W \neq 0$.

If cost information is unavailable for a material share of sold units, the system reports sales and cost coverage rather than silently interpreting revenue as profit. Average order value uses eligible completed orders as its denominator; repeat-rate reporting states customer-identification coverage; and stock-turnover reporting specifies both the cost and average-inventory conventions used. Period comparisons use comparable windows or explicitly identify differences in trading days and closures.

Each KPI response therefore contains a value, definition version, reporting period, aggregation level, coverage indicators, and reconciliation status. This structure makes changes in business-rule definitions visible and allows the evaluation to distinguish calculation anomalies from genuine changes in business performance.

3.3.2 Demand Prediction

The implemented RQ1 target is observed units at shop–category–day level. For series s, origin t and horizon $h \in \{1, 7, 14, 28\}$, the direct cumulative target is

$$Y_{s,t,h} = \sum_{j=0}^{h-1} y_{s,t+j}.$$

Separate models are fitted for each horizon. Features must be available strictly before t, and every training target window must end before its validation origin.

Historical features include lags 1, 7, 14, 28 and 56; rolling mean, standard deviation, sum and zero ratio over 7, 14, 28 and 56 days; days since last positive sale; ADI; positive-demand mean and CV^2; shop and category identifiers; and calendar encodings. The context-aware challenger adds past shop, category and network aggregates, recent trends and the series' share of those aggregates. Context features use only observations before the origin.

Candidates include seasonal-naive, Croston–SBA, TSB, LightGBM Tweedie, CatBoost Tweedie and a two-stage occurrence–quantity model. ADI and CV^2 support diagnosis but do not automatically route a model. Specialist methods are promoted only with adequate common-origin validation and independent series support.

The forecast estimates observed sales, not unconstrained latent demand. Stock-out censoring, promotion effects and SKU-level forecasts require additional point-in-time evidence and separate validation.

3.3.3 Adaptive Forecasting Architecture and Promotion Gates

The architecture separates source eligibility, target construction, candidate forecasts, model selection and business action. Source and target gates distinguish observed zero from censored or unknown demand. Candidates are compared on common rolling origins, bias and independent-series support. Uncertainty and aggregation follow selection; inventory actions require validated costs, lead time, stock and owner constraints.

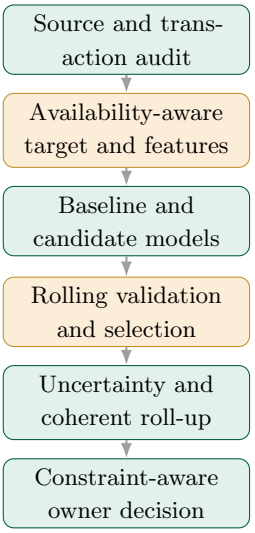


FIGURE 3.1: Evidence-gated adaptive demand-forecasting architecture.

3.3.4 Future Customer Inactivity

For customer c at historical cutoff t, future inactivity is represented as

$$q_{c,t} = P(\text{no purchase in } (t, t+H] \mid X_{c,t}), \tag{3.4}$$

where $X_{c,t}$ contains only records available by cutoff t, and H is the predefined future outcome horizon. The value of H will be fixed before final evaluation with reference to the relevant business purchase cycle.

The resulting $q_{c,t}$ represents predicted inactivity risk and does not, by itself, estimate the causal effect of contacting customer c.

3.3.5 Strategy Optimization

For an eligible action $a \in A_t$, the system evaluates

$$U(a \mid D_t) = \mathbb{E}[\Delta\text{Profit}(a)] - \text{ActionCost}(a) - \lambda\,\text{Risk}(a), \tag{3.5}$$

subject to operational constraints including budget, storage, minimum-order quantity, supplier lead time, pack multiple, remaining shelf life, expiry, temperature-controlled storage capacity, incoming stock, available inventory, and customer-contact consent where applicable.

This transparent objective represents the transition from predictive outputs to prescriptive decisions. Its terms will be estimated using available evidence, and uncertain values will be examined through sensitivity analysis rather than treated as precisely known constants.

3.3.5.1 Inventory Position and Replenishment Cost

Inventory position combines usable on-hand stock and confirmed incoming quantity, less committed demand. Replenishment is constrained by supplier lead time, pack multiple, minimum order quantity, budget, storage, expiry and remaining shelf life. An action is withheld when a required field is unknown or a hard constraint is violated.

Policy evaluation compares the owner policy and proposed policy on the same demand path and opening inventory. Outcomes include stock-out units, average inventory, holding cost, shortage cost, expiry and total cost. Forecast error alone is not interpreted as inventory benefit.

3.3.5.2 Customer Action Logic and Abstention

Customer-action logic differs from replenishment logic. The inactivity probability $q_{c,t}$ in Equation 3.4 may be used to prioritize customers for review, but it does not estimate the causal effect of contacting customer c.

A contact action is eligible only when customer consent and business policy permit contact, the proposed treatment is specified, and sufficient operational capacity exists. Expected profit will be calculated only when a defensible estimate of incremental response benefit is available. Otherwise, the output will explicitly remain a risk-ranked contact list. This separation prevents a predictive probability from being incorrectly interpreted as campaign uplift.

The action layer may also abstain from making a recommendation. Abstention is appropriate when required constraints are missing, uncertainty makes the leading alternatives practically indistinguishable, predicted benefit does not exceed action cost over the declared sensitivity range, or the relevant model fails a monitoring gate. An abstention accompanied by a reason is therefore treated as valid safety behaviour rather than as a system error.

3.3.6 Preprocessing and Historical Snapshots

The processing sequence is fixed: validate the source and schema; remove or flag exact duplicates; reconcile cancellations and returns; construct the data split; fit imputation, encoding, and scaling transformations using training data only; construct features; train candidate models; and evaluate them on the untouched test set. This ordering prevents information from future periods from leaking into model development [15].

Demand features may include lagged demand, rolling location and variability statistics, day of week, month, distance to relevant festivals, price, discount, known promotions, stock-out indicators, pack units, remaining shelf life, and external market context. Batch-expiry and temperature-controlled storage variables primarily constrain action feasibility and enter prediction only when their decision-time availability and analytical role are defensible.

Where feasible, zero observed sales caused by zero demand will be distinguished from zero sales caused by lack of stock. Intermittency will be measured before selecting temporal granularity and candidate forecasting methods.

Customer snapshots may include RFM variables, tenure, order-value statistics, category breadth, purchase intervals, discount behaviour, and pre-cutoff return rates. The outcome label is constructed from a subsequent future window. Where required, a purge interval will separate the observation and label windows to prevent overlap or delayed events from introducing leakage.

Customer identifiers used for an entity-generalization test will not appear on both sides of the corresponding training–test boundary.

3.4 Candidate Methods and Selection Rules

3.4.1 Baseline Ladder and Task-Specific Candidates

Method selection follows a baseline ladder, with model eligibility determined by the available evidence volume and task structure.

Performance: Auditable KPI definitions and rule-based anomaly detection provide the descriptive baseline.

Demand: Last-value, weekly seasonal-naive, moving-average, and exponential-smoothing baselines are evaluated before ARIMA/SARIMA, Croston–SBA, TSB, and gradient-boosted lag models. LightGBM and CatBoost are candidates for pooled shop/category or SKU features when the sample and categorical structure support them. LSTM models are considered only when sequence volume and history justify their additional complexity. Intermittent-demand methods are promoted only after same-origin validation [6, 24, 26, 27].

Inactivity: A recency or base-rate rule is compared with Logistic Regression, Random Forest [2], and XGBoost [5]. Class weighting and decision thresholds are selected using development data only.

Segmentation: Appropriately scaled RFM features are analysed using K-Means. Candidate values of K are evaluated using inertia, silhouette, resampling stability, and business interpretability.

Explanation: Coefficient- or rule-based explanations are preferred for simple models, while SHAP provides model-behaviour attributions for compatible tree-based models [16]. Displayed explanations will remain selective and context-specific [18].

3.4.2 Model Eligibility and Sector-Aware SME Adaptation

Candidates are compared only when target, origin, horizon, training information and evaluation rows are aligned. Every task begins with a baseline. Pooled models may borrow information across businesses; business adaptation requires sufficient independent history.

Selection considers error, aggregate bias, calibration, stability across origins and independent-series support. Repeated forecasts of one series do not create independent businesses. Regime labels support diagnosis but do not justify automatic specialist routing when subgroup support is sparse. Sector-specific extensions are enabled only when relevant fields and operating rules are available.

3.5 The B-SMART Sequential Procedure

3.5.1 Strategy Recommendation Algorithm

FIGURE 3.2: B-SMART Strategy Recommendation

Input: Data available at cutoff D_t, horizon H, constraints C_t, validated model/baseline versions M, and candidate actions A

Output: Ranked feasible recommendations R_t

Check sources, schema, timestamps, and organizational ownership
Create the historical feature snapshot X_t
Apply transformations fitted on the appropriate training evidence
Compare eligible baseline, pooled-business, and shop-adapted candidates
Estimate demand uncertainty and customer inactivity probabilities
Compute reproducible KPIs and data-quality alerts
Generate candidate actions with source and decision-time metadata
for each $a \in A$ **do**
Check budget, stock, pack, applicable shelf-life, storage, lead-time, capacity, and consent feasibility under C_t
Estimate benefit, cost, and uncertainty using declared assumptions
$U(a) \leftarrow \mathbb{E}[\Delta\text{Profit}(a)] - \text{Cost}(a) - \lambda\,\text{Risk}(a)$ **end for**
Remove infeasible actions and rank the remaining actions by $U(a)$
Attach a concise explanation, uncertainty statement, cutoff, and evidence source
Return the top-K actions and record the owner's response and any subsequent observable outcome

3.5.2 Computational Complexity

For N invoice lines, d validated fields, and P candidate actions, source validation and feature aggregation are approximately

$$O(Nd), \tag{3.6}$$

while ranking the candidate actions requires approximately

$$O(P \log P). \tag{3.7}$$

The computational complexity of model training depends on the selected model and will be measured empirically. Serving uses versioned artifacts and therefore does not retrain the model within the request path.

3.6 Proposed Architecture and Information Flow

3.6.1 Architecture Components

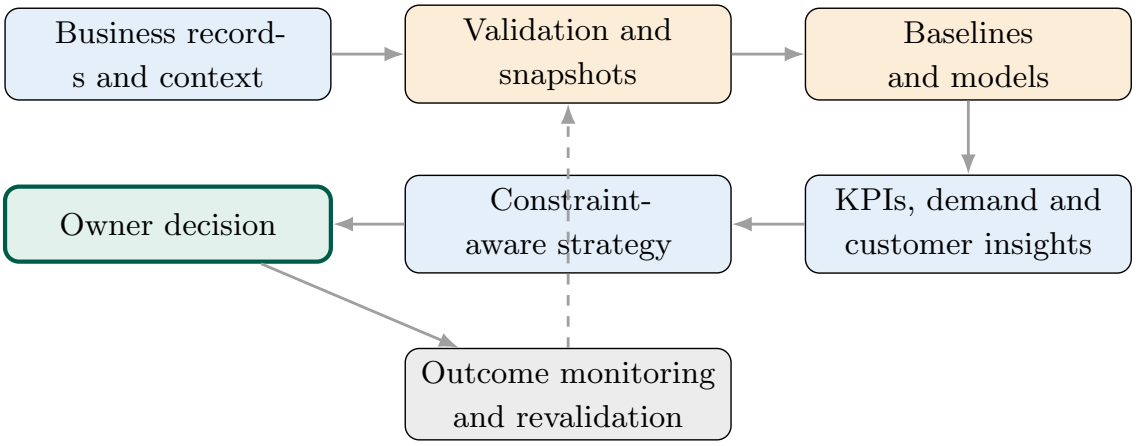


FIGURE 3.3: System information flow and owner oversight.

The operational layer preserves business boundaries and source lineage. The analytical layer compares baselines and eligible models, then produces KPIs, forecasts and customer insights. The strategy layer combines those outputs with business constraints before displaying a recommendation. Owners remain decision-makers. Outcomes enter monitoring and become training evidence only after reconciliation and validation for a later model version.

3.6.2 Monitoring and Continuous Learning

Information flow is deliberately one-way at prediction time: future outcomes cannot alter the historical snapshot that generated an earlier prediction or recommendation.

Operational state and recommendations may be updated when relevant events occur or according to a defined schedule. However, the deployed model is not blindly retrained after every transaction.

Observed outcomes enter the monitoring process and become eligible training evidence only in a later model version after reconciliation, label maturation, drift assessment, and comparison with the existing model and baseline.

This monitored feedback lifecycle represents the intended meaning of continuous learning while preserving the ability to reproduce what the system knew when a particular action was proposed.

3.7 Snapshot Lineage and Frozen Evaluation Flow

3.7.1 Historical Snapshot Construction

Each snapshot is defined by cutoff t. Only records available by that cutoff may enter the prediction. Event time and recording time are retained separately so late-arriving or corrected records cannot silently change a past information set.

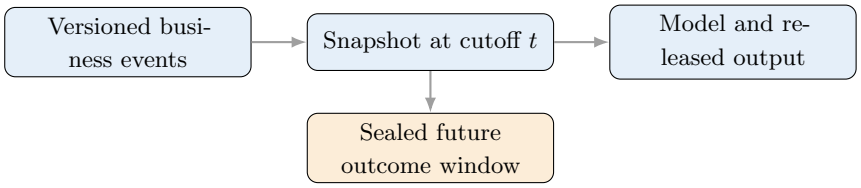


FIGURE 3.4: Point-in-time snapshot and future-outcome separation.

For forecasting, each training target window ends before its validation origin, and transformations are fitted within the training window. Customer labels are assigned only when the complete future horizon is observable. Overlapping outcomes require an appropriate purge interval.

3.7.2 Frozen Final Evaluation and Outcome Maturation

Training and validation snapshots are used for model development. The final test set remains sealed until the task definition, feature and eligibility rules, candidate models, primary metrics, decision thresholds, and random seeds have been frozen.

If a defect is identified after the final test has been opened, the defect and its impact will be documented. The affected observations cannot subsequently be treated as though they were still part of an untouched final test. This rule protects the interpretation of the final result rather than merely enforcing the mechanics of a data split.

Outcome records are not automatically incorporated into a released model. Observed outcomes first enter the monitoring store. They become eligible for a later model version only after reconciliation, label maturation, and repetition of the appropriate validation protocol.

This delay ensures that an owner's response, a corrected invoice, or a partially observed outcome cannot retrospectively change the evidence that supported a historical production decision.

3.8 Owner-Facing Recommendation Evidence Contract

3.8.1 Minimum Recommendation Evidence

Each recommendation identifies its business, decision, target, cutoff, model or rule version, uncertainty and supporting evidence. Reorder suggestions expose current and incoming stock, forecast horizon, lead time, pack requirements, budget and expiry risk. Customer actions expose the historical behaviour supporting the priority, probability reliability and consent status.

When a required field is missing, the system withholds the action or labels the result incomplete. It does not assume costs, stock or customer responses. Owners may accept, reject, modify or defer actions; responses and later observable outcomes are retained for audit.

3.8.2 Human Use and Accountability

An explanation is considered successful only when it contributes to correct and appropriate use. The owner study will therefore include scenarios in which the appropriate decision is to reject or defer an apparently attractive recommendation.

This evaluates calibrated human reliance rather than merely whether participants prefer a more detailed interface. The evidence contract also supports failure analysis because an inappropriate decision can be traced to the underlying estimate, missing constraint, explanation, outdated evidence, or owner interpretation rather than being attributed generically to the "AI."

3.9 Evaluation Protocol

Evaluation follows development, protocol freeze, untouched testing and bounded reporting. Candidates, features, thresholds and primary metrics are selected using development data only. The final test is opened after the protocol is frozen; later changes are exploratory.



FIGURE 3.5: Frozen evaluation sequence.

3.9.1 Forecasting Evaluation

All forecasting splits preserve temporal order. Candidate methods are selected using rolling-origin or expanding-window validation, followed by one evaluation on a final untouched period.

For observed values y_i and forecasts $\hat{y}_i$, the principal error measures include

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|\,, \tag{3.8}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \tag{3.9}$$

$$\mathrm{WAPE} = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{\sum_{i=1}^{n} |y_i|}, \qquad \sum_{i=1}^{n} |y_i| > 0. \tag{3.10}$$

MASE will additionally support scale-independent comparison and avoids several limitations of conventional percentage errors [13]. For the current cumulative-horizon benchmark, the reported horizon-scaled weekly MASE uses the per-series mean absolute lag-7 difference multiplied by the forecast horizon. Cluster bootstrap resampling is performed at the shop–category-series level to preserve repeated-origin structure. Prediction-interval coverage and interval width will be used to evaluate uncertainty.

Results will be stratified by demand volume and intermittency so that poor performance on low-volume or sparse products is not hidden inside an aggregate average.

3.9.2 Customer and Segmentation Evaluation

Customer-inactivity models will report outcome prevalence, PR-AUC, ROC-AUC, Lift@10%, Brier score, and probability-reliability analysis. Precision–recall measures are emphasized under class imbalance [21], while calibration is evaluated independently because a model may produce useful rankings while still producing misleading probabilities [19].

Decision thresholds and contact fractions will be selected using validation evidence and declared cost assumptions. They will not be modified after the final test outcomes have been inspected.

Clustering evaluation will report inertia, silhouette, cluster stability, minimum cluster size, and corresponding RFM profiles. Final segment interpretations will also be reviewed with domain participants.

If separation is weak or unstable, the result will be reported as such. The existence of algorithmically generated clusters will not be treated as evidence that natural or causal customer types exist.

3.9.3 Ablation, Robustness, and Statistical Analysis

Planned ablations include local-calendar features, price and promotion features, market context, pack and expiry context, and customer-history extensions.

Comparisons will include the transparent baseline, eligible pooled-business models, and shop-adapted models where the available evidence permits them. Where augmentation is used, real-only and augmented variants will be evaluated under controlled comparisons.

Results will be presented by business, with paired comparisons conducted over common forecast origins or comparable analytical instances. A held-out-business transfer test will be used where sufficient participating businesses are available.

Uncertainty will be summarized using paired block-bootstrap intervals across time or business units, depending on the relevant dependence structure. Differences between models will be reported using effect estimates and uncertainty intervals rather than significance labels alone.

Sensitivity analysis will vary supplier lead time, shortage cost, holding cost, action cost, and the action-risk penalty used in Equation 3.5.

3.9.4 Frozen Analysis and Statistical Decision Rules

One primary comparison and metric per research question are frozen before the final test. RQ1 uses paired out-of-time errors on common origins; RQ2 uses PR-AUC with calibration as a separate criterion; RQ3 compares policies on identical demand paths and constraints; and RQ4 evaluates task correctness under counterbalanced owner tasks.

Uncertainty methods must respect dependence: neighbouring dates, repeated customer snapshots and observations from one business are not independent. Paired block-bootstrap or cluster methods are selected according to the unit of analysis. Effects are reported with intervals and interpreted for practical importance, not significance labels alone.

3.9.5 Practical Importance and Missing Outcomes

Practical importance will be evaluated independently of statistical evidence. A small reduction in average forecasting error may be operationally important for a high-volume product but irrelevant after ordering constraints are applied. Similarly, a moderate increase in task-completion time may be acceptable if it produces a material improvement in decision correctness.

Where defensible cost or effect thresholds can be established with participating businesses, they will be specified before the final comparison. Where no defensible threshold exists, the magnitude and uncertainty of the effect will be reported without assigning an arbitrary success label.

Secondary metrics, alternative horizons, customer segments, and feature interactions remain useful for diagnosis but will be identified as secondary or exploratory analyses. When many related comparisons are examined, the full set of relevant results will be reported rather than only favourable cases.

Any post-test change to a model, threshold, feature definition, or analysis creates a new exploratory version rather than replacing the original primary result. This separation allows continued model improvement without rewriting the original evidence.

Missing outcomes will be explicitly identified. An incomplete future horizon cannot automatically be treated as evidence of customer inactivity. An unknown inventory balance cannot be treated as zero stock. Similarly, a recommendation without a later observable outcome may support a usability analysis but cannot support a claim of realized business impact.

Where assumptions can reasonably be bounded, sensitivity analysis will be used. Otherwise, the corresponding claim will be classified as not evaluable with the available evidence.

3.9.6 Decision, Usability, and Scalability Evaluation

Replenishment policies are compared on identical demand paths and opening inventory using stock-out units, average inventory, holding cost, shortage cost, expiry and total cost. Retention value is estimated only when incremental response, contact cost and consent evidence exist; otherwise the task remains prioritisation, not realized-profit estimation.

Owner evaluation measures decision correctness, completion time, confidence, error count and usability. Generated test records may support software-load and scalability checks, but are not mixed with empirical forecast or business-impact results.

3.9.7 Monitoring Plan

Monitoring will distinguish among schema or data-quality failure, input distribution shift, calibration change, and outcome-performance deterioration.

An alert does not automatically authorize retraining. Any replacement model must pass the validation procedure again and retain its training window, feature version, parameters, and comparison results.

This requirement ensures that adaptation is evaluated before deployment rather than assumed to be beneficial merely because new data have become available.

3.10 Reproducibility, Audit, and Failure Reporting

3.10.1 Reproducibility Levels

Four levels of reproducibility will be designed.

Data reproducibility: requires a source manifest, checksum, schema version, mapping rules, and exclusion log.

Analytical reproducibility: requires snapshot identifiers, fitted transformations, feature definitions, target construction, and split manifests.

Model reproducibility: requires code, execution environment, parameters, random seeds, trained artifacts, and the evaluation command.

Decision reproducibility: additionally requires the applicable constraints, cost assumptions, candidate-action set, ranking rule, and interface version.

3.10.2 Restricted and Shareable Evidence

Primary partner data do not need to be publicly released in order for the study to maintain reproducibility controls. The reproducibility package will therefore distinguish restricted evidence from shareable artifacts.

Shareable materials may include schemas, data dictionaries, manifest templates, transformation code, generated test fixtures, and configuration files without revealing confidential identifiers or transaction records.

Where restricted evidence is available to an authorized evaluator, the available package should permit that evaluator to reconstruct the relevant snapshot and result. An external reader should additionally be able to inspect the process using a public or generated test fixture.

3.10.3 Experiment Records and Correction History

Each experiment will generate a machine-readable run record containing, where applicable, the research question, dataset and snapshot hashes, training and test windows, grouping keys, feature version, candidate and baseline versions, parameters, output-artifact hashes, warnings, and start and finish times.

Manual spreadsheet corrections will not be made outside this record. If an owner supplies a correction, it becomes a dated source or mapping revision. The correction affects downstream outputs only when a new version is executed.

3.10.4 Failure Reporting

Failure reporting is part of the scientific output. Data-eligibility failure, lack of convergence, numerical instability, training failure, resource exhaustion, and constraint infeasibility will be distinguished from ordinary evaluation outcomes.

The number of attempted cases will be reported. Results will not be summarized only over successful model runs while silently excluding failures.

A method that fails on sparse or short series will therefore be evaluated against the full eligible set to which it was applied, rather than only the subset for which it returned a prediction.

3.10.5 Recommendation Auditability and Claim Traceability

Auditability extends to the owner interface. Every recommendation receives an identifier linking the evidence contract, displayed recommendation, owner response, and any later observable outcome.

If a recommendation changes following a data correction or model replacement, the previous record remains immutable and the successor record refers to it. This supports accountability without implying that the model, researcher, or owner alone caused the eventual business outcome.

The final research package will also include a claim-to-evidence index. Each central claim will identify the corresponding research question, figure or table, experimental record, and evidence tier.

Claims supported only by public, generated, simulated, or small-sample evidence will remain explicitly labelled as such. This index provides a final safeguard against expanding the scope of a claim beyond what was actually evaluated.

3.11 Threats to Validity and Planned Controls

The main threats are source-field errors, incomplete inventory exposure, business heterogeneity, temporal leakage, sparse demand, uncertain costs and limited independent businesses. Controls include transaction reconciliation, task eligibility, point-in-time features, common-origin baselines, cluster-aware uncertainty, sensitivity analysis and an untouched final test.

The supplied records support the current shop–category observed-sales experiment. They do not alone establish SKU latent demand, causal promotion effects, inventory-policy savings or transfer to every SME sector. Those claims require additional validated fields or separate participant-based evaluation. Unavailable evidence is reported as a limitation.

3.12 Expected Methodological Outputs

At completion, the methodology is expected to produce a documented canonical business schema and data dictionary, a versioned feature and partner-data protocol, and explicit specifications for transparent baselines, pooled-business models, and eligible shop-adapted models.

The study will also produce a reproducible evaluation package, a constraint-aware recommendation procedure, and an owner-study instrument.

Effectiveness claims will be restricted to the sampled businesses, sectors, tasks and periods actually evaluated. Transfer to a new sector requires schema and constraint mapping, new evidence and held-out-sector evaluation.

The system supports business decisions under owner oversight; automated actions outside the evaluated business tasks remain excluded.

Implementation details, observed numerical results, and conclusions will be reported only when the corresponding empirical evidence is available.

Chapter 4

Implementation

4.1 Dataset Package and Research Scope

The supplied dataset manifest claims 100% real SME sales and real people, with Bangladesh as the business context, BDT currency and coverage from 1 January 2023 to 31 December 2025. It lists 32 shops, 11,000 customers, 480 products and 56 suppliers. The source package includes 365,447 sales orders, 736,683 sales lines, 381,756 payments, 122,422 purchase orders and purchase lines, 994,808 inventory movements, 134,379 inventory lots, 5,102 returns, 1,216 promotions and 1,096 calendar rows.

The same manifest records a generation seed and describes business validation as a "modeled world" for Bangladeshi SMEs. The RQ3 future-demand field uses generated scenario demand, and RQ4 consists of prepared rule-test cases rather than participant responses. No independent source-system export, business-owner attestation or source lineage was available in this run. The package is therefore treated as researcher-supplied data with unverified provenance; internal joins establish consistency, not real-world origin or representativeness.

The RQ1 experiment uses the dedicated complete daily shop–category panel: 32 shops, 16 categories, 512 series and 1,096 daily observations per series, for 561,152 rows. The panel contains 1,580,361 observed units. The separate simulated unmet-demand table is not used as observed sales or as empirical lost-sales ground truth. Raw business records remain in controlled storage; the public repository contains code, schemas, checksums and aggregate outputs.

TABLE 4.1: Source-table coverage declared in the supplied dataset manifest.

Source table	Rows	Analytical role	Use in current RQ1
Shops	32	Business and shop identifiers	Series grouping
Customers	11,000	Customer-level operational history	Not used as a forecast target
Products	480	SKU and category master	Category mapping and reconciliation
Suppliers	56	Supplier and procurement context	Not yet used as a predictor
Sales orders	365,447	Order headers and shop/customer links	Parent transaction validation
Sales lines	736,683	SKU-level quantities and values	Source reconciliation and aggregation
Payments	381,756	Payment events and settlement evidence	Not used as a forecast predictor
Purchase orders / lines	122,422 / 122,422	Procurement and received quantities	Inventory reconciliation
Inventory movements	994,808	Opening, purchase, sale and adjustment events	Stock-exposure diagnostics
Inventory lots	134,379	Lot and batch traceability	SKU eligibility diagnostics
Returns	5,102	Reversal and return events	Reconciliation context
Promotions	1,216	Promotion records	Not used until point-in-time approval is validated
Calendar	1,096	Daily calendar coverage	Calendar feature validation

The table distinguishes counts declared by the manifest from fields actually used by RQ1: the presence of a source table does not make every field an eligible predictor. A variable is admitted only after its join key, business meaning and availability time have been checked. The manifest's real-data claim remains unverified.

4.2 Data Validation and Preprocessing

The ingestion procedure parses the date field, checks required columns and missing values, rejects duplicate shop–category–date keys, verifies that units sold are finite and non-negative, and checks that the zero-sales flag agrees with the observed quantity. It then verifies that every shop–category series contains the same complete daily calendar. No missing dates are filled by assumption. The audit found zero duplicate keys, zero missing required values, zero zero-flag mismatches, and zero missing dates in the supplied panel. The zero-sales share is 37.5962%.

The dedicated SKU-level export is not substituted for this complete shop–category panel. Its absent SKU-date rows require a separate source-aware decision about assortment, shop opening, product activation, stock-outs, and export completeness before zero demand can be assigned. This distinction prevents the conversion of unobserved records into artificial zero labels.

Table 4.2: Automated integrity checks on the complete shop–category panel.

Check	**Observed result**	**Interpretation**
Panel rows	561,152	512 series multiplied by 1,096 dates
Duplicate shop–category–date keys	0	Key uniqueness passed
Missing required values	0	Required panel fields complete
Zero-flag / quantity mismatches	0	Boolean zero indicator agrees with quantity
Missing daily dates within series	0	Complete daily panel; no date imputation required
Observed units	1,580,361	Total target quantity in the panel
Explicit zero-sales rows	210,972	37.5962% of panel observations

The zero-sales flag is checked against the observed quantity rather than trusted as an independent label. A row is an observed zero only because the complete shop–category panel explicitly contains that date and quantity. This rule must not be transferred automatically to the sparse SKU export, where an absent row may mean an inactive product, stock-out, closed shop, incomplete export or genuinely unobserved period.

4.3 Transaction and Panel Reconciliation

The 736,683 sales lines join to 365,447 sales orders without missing order references or shop mismatches. Aggregating source sales lines by `shop_id`, `sku_id` and calendar date reproduces all 699,220 rows of the prepared positive-sale SKU table exactly, including 1,580,361 units and the recorded sales value to floating-point precision. This is an internal reconciliation of the supplied tables. It confirms record-level consistency across linked tables, but does not independently establish where the records originated.

The prepared SKU export contains 32 shops, 480 SKUs and 13,271 shop–SKU combinations with at least one positive-sale record. It contains only positive-sale days; its `zero_day_interpretation` field warns that missing SKU dates are not automatically zero demand. The complete shop–category panel contains five fields: `shop_id`, `category`, `date`, `units_sold_observed`, and `is_zero_observed_sales`. It has 561,152 rows, 32 shops, 16 categories, 512 series and 1,096 consecutive daily observations per series. The total is 1,580,361 units and 210,972 rows (37.5962%) are zero-sales observations. Aggregating SKU sales to shop–category–date reproduces all 350,180 positive category-day rows exactly; the remaining 210,972 category rows are explicit zero-sales rows in the complete panel.

Table 4.3: Cross-table reconciliation results and unresolved exceptions.

Reconciliation	Result	Research implication
Sales lines to order headers	736,683 lines join to 365,447 orders with no missing order reference or shop mismatch	Parent transaction linkage passed
SKU daily aggregation	699,220 positive-sale rows and 1,580,361 units reproduced from source lines	Prepared SKU aggregate is internally reproducible
Shop–category aggregation	350,180 positive category-days reproduced; remaining 210,972 rows are explicit zero observations	Complete category panel reconciles to SKU sales
Sale inventory movements	All 736,683 match sales lines by order, shop, SKU, lot and quantity	Sale ledger linkage passed
Purchase movements	121,107 movement groups match received purchase-order/SKU groups	Received procurement quantities reconcile
Pending purchase lines	1,315 have no final received quantity at cutoff	Remain pending; not counted as available stock

4.4 Inventory and Procurement Reconciliation

The inventory ledger contains 13,272 opening movements, 121,107 purchase movements, 736,683 sale movements, 118,673 expiry-disposal movements and 5,073 damage movements. Every sale movement matches a sales line by order, shop, SKU, lot and quantity. The 121,107 purchase movement groups match received purchase-order/SKU quantities; 1,315 purchase lines have no final received quantity and remain pending at cutoff.

A chronological stock reconstruction identifies 269 negative-balance movement events across 253 shop–SKU combinations. It also identifies 330 positive-sale days with non-positive reconstructed intraday stock. These records are quarantined from SKU exposure labels pending correction. Across the 13,272 inventory-tracked shop–SKU combinations and 1,096 dates, the ledger reconstructs 13,781,575 fully in-stock days and 764,537 days with non-positive intraday balance. These are derived diagnostics, not an independent stock audit. A separate exposure-aware panel run quarantined all 253 shop–SKU histories containing negative balances. The resulting panel contains 13,545,871 eligible shop–SKU–days: 685,827 positive-sale days and 12,860,044 zeros created only where reconstructed

intraday stock remained positive. The run excluded 13,393 positive-sale records from exposure labels because their histories were quarantined or their reconstructed intraday balance was non-positive. Every zero-sales shop–category day has at least one recorded SKU with positive reconstructed stock during that day; this does not rule out item-level stock-outs or unrecorded assortment.

Table 4.4: Inventory movement composition and reconstructed balance diagnostics.

Movement type	**Rows**	**Treatment**
Opening	13,272	Initial recorded stock position
Purchase	121,107	Reconciled to received purchase-order/SKU groups
Sale	736,683	Matched to sales lines
Expiry disposal	118,673	Removed from usable inventory
Damage	5,073	Removed from usable inventory
Total	**994,808**	Ledger movement count

Table 4.5: Stock reconstruction exceptions requiring quarantine before SKU target construction.

Diagnostic	**Count**	**Interpretation**
Negative-balance movement events	269	Across 253 shop–SKU combinations; investigate ledger sequence or opening balance
Positive-sale days with non-positive intraday minimum stock	330	Cannot be treated as fully in-stock exposure without correction
Reconstructed fully in-stock days	13,781,575	Derived ledger diagnostic across 13,272 tracked shop–SKU combinations
Days with non-positive intraday balance	764,537	Exposure is not eligible as a known positive-stock day
Eligible shop–SKU–days after full-series quarantine	13,545,871	Zeros assigned only within reconstructed positive-stock exposure
Positive-sale days excluded from exposure labels	13,393	Includes quarantined shop–SKU histories and non-positive intraday stock

The reconstruction is a derived diagnostic, not an independent physical stock count. Every zero-sales shop–category day has at least one recorded SKU with positive reconstructed stock during that day, but this does not rule out item-level stock-outs or unrecorded assortment. The 2,088 theoretical shop–SKU combinations without an opening movement are not assumed to be active series. The exposure-conditioned SKU experiment in Section 5.11 uses the reconstructed panel as a diagnostic; physical availability and the Q4 period are not independently verified or untouched.

4.5 RQ1 Forecasting Pipeline

For each shop–category series, the feature builder uses only information strictly before the forecast origin. Features include lags of 1, 7, 14, 28, and 56 days; rolling mean, standard deviation, sum, and zero-sales ratio over 7, 14, 28, and 56 days; days since the last positive sale; ADI; positive demand mean and CV^2; shop and category identifiers; weekday, day, month, ISO week, and cyclical calendar encodings. No future observed sales are used as predictors.

Four direct cumulative targets are evaluated: 1, 7, 14, and 28 days. Development forecast origins are 1 July, 1 August, and 1 September 2025. The previously inspected fourth-quarter 2025 period is not used for model selection. Historical training examples are sampled every 28 days; each training label window must finish before the corresponding validation origin. A 56-day history is required before a feature row is admitted.

The LightGBM Tweedie candidate uses variance power 1.3, 100 boosting iterations, learning rate 0.035, 31 leaves, minimum child samples 80, row and column subsampling 0.85, subsampling frequency 1, L2 regularisation 2.0 and seed 42. The CatBoost Tweedie challenger uses 120 iterations, depth 6, learning rate 0.04 and L2 regularisation 5. The two-stage CatBoost hurdle uses a 100-iteration occurrence classifier and a 100-iteration positive-quantity regressor, depth 5 and learning rate 0.04; the classifier uses balanced class weights. Its probabilities are not calibrated and its output is not selected for deployment. Croston–SBA and TSB use $\alpha = 0.1$ (and $\beta = 0.1$ for TSB). The weekly seasonal-naive baseline averages up to four previous observations for each matching weekday and sums them across the horizon. All methods are evaluated on the same 512 series at each of the three origins, giving 1,536 series-origin forecasts per horizon.

Table 4.6: Feature groups and point-in-time availability rules for the RQ1 model.

Feature group	**Variables**	**Availability rule**
Own-series lags	1, 7, 14, 28 and 56 days	Observations strictly before forecast origin
Rolling demand	Mean, standard deviation, sum and zero ratio over 7, 14, 28 and 56 days	Window ends at origin minus one day
Intermittency	Days since last sale, ADI, positive-demand mean and CV^2	Calculated from pre-origin history only
Calendar	Weekday, day, month, ISO week and cyclical encodings	Calendar known at the forecast origin
Cross-series context	Shop, category and network totals; trends; series shares	Aggregates use only pre-origin observations
Potential business covariates	Price, discount, promotion, supplier and stock context	Excluded unless point-in-time availability and joins are validated

The feature policy prevents target leakage at two levels. First, no feature window includes the date being forecast or any later date. Second, a historical training example is admitted

only when its entire cumulative target window ends before the validation origin. The same origin, target definition and eligible rows are used for baseline and context comparisons.

4.6 Context-Aware Forecasting Model Development

The existing pooled LightGBM Tweedie model was retained as the controlled baseline. The challenger used the same estimator family and settings while adding cross-series information available before each forecast origin: shop-level sums and means, category-level sums and means, network totals, recent shop/category trends and the focal series' share of its shop, category and network totals. This tests whether related series provide useful context without introducing future information.

The seven-day comparison uses nine monthly origins from January to September 2025, with all 512 series evaluated at every origin. One-, 14- and 28-day comparisons use three origins from July to September 2025. Q4 2025, which had already been inspected, is not used for model selection. Every training target window ends before its validation origin. Baseline and context models use the same target, origins and LightGBM settings.

The model remains a shop–category observed-sales forecaster. It does not convert missing SKU dates to zero or claim to estimate unconstrained latent demand. Code and aggregate outputs are stored under `research/forecasting/context_aware_experiment.py` and `research/forecasting/results/context_aware_v1/`.

Table 4.7: LightGBM Tweedie configuration shared by the baseline and context challenger.

Parameter	**Value**	**Purpose**
Objective / variance power	Tweedie / 1.3	Non-negative, right-skewed demand target
Boosting iterations	100	Fixed candidate complexity
Learning rate	0.035	Step size for boosting
Leaves / minimum child samples	31 / 80	Tree complexity and leaf support
Row / column subsampling	0.85 / 0.85	Stochastic regularisation
Subsample frequency	1	Apply row subsampling each iteration
L2 regularisation	2.0	Penalise excessive leaf weights
Random seed	42	Reproducibility of stochastic steps

The context challenger changes the information set, not the estimator family or hyperparameter budget. This controlled design makes the feature-group comparison easier to interpret: an observed difference is associated with the added historical context rather than a simultaneous change to the algorithm and its tuning configuration.

4.7 Historical Transaction-Feature Ablation

A separate seven-day ablation tests whether transaction-derived covariates reduce error beyond the historical demand features. Sales lines are joined to parent orders and the product master, then aggregated to shop–category–day for gross value, discount, order count, customer-days and quantity-weighted unit price. The aggregated quantity reproduces the complete observed shop–category panel exactly. The extended feature set uses only historical price, discount, order and customer-activity summaries from the 7-, 28- and 56-day windows before each forecast origin. It does not use future prices, promotion schedules, inventory outcomes or customer responses.

The ablation compares baseline LightGBM Tweedie with the same model plus these transaction-derived features at nine monthly origins from January through September 2025. Each origin covers all 512 series. Training windows use a 28-day stride and a minimum 56-day history; every training target window ends before its validation origin. The script and aggregate outputs are stored in `research/forecasting/ablation_transaction_features.py` and `research/forecasting/results/transaction_feature_ablation_v1/`. The ablation is development evidence, not an untouched final test.

4.8 Software Environment and Reproducibility

The experiment was executed with Python 3.13.5, pandas 2.2.3, NumPy 2.3.5, scikit-learn 1.8.0, and LightGBM 4.6.0. The source panel's SHA-256 checksum is recorded in the accompanying experiment manifest. The implementation writes the per-origin metrics, aggregate metrics, forecast-level predictions, demand-regime assignments, and regime-stratified metrics as machine-readable CSV files. The raw business dataset is retained in its controlled data location and is not copied into the public source repository.

4.9 Current Implementation Boundaries

The multi-horizon LightGBM experiment remains a shop–category observed-sales forecast. A separate SKU diagnostic uses reconstructed exposure and reports the 28-day exposed-sales mean as the lowest-WAPE candidate; its Q4 2025 period had already been inspected and is not an untouched final test. CatBoost Tweedie and a raw two-stage hurdle remain development challengers; hurdle probabilities are uncalibrated. The RQ2 model comparison is supplemented by a six-date rolling ranking validation. RQ3 includes the supplied scenario analysis and an observed-sales historical backtest, but no actual owner-policy replay or audited accounting cost. RQ4 has only a computational case audit. No participant usability study has been completed. The data manifest's real-origin claim remains unverified. A future holdout, independent business provenance, owner-policy replay and consented user study remain necessary before field-effectiveness claims.

4.10 RQ2 Customer Inactivity Pipeline

The supplied 90-day inactivity label was independently reconstructed from the supplied sales-order dates: a snapshot is labelled inactive when no customer order occurs after its cutoff and on or before its 90-day horizon end. All 72,146 labels matched this rule. The supplied temporal split was preserved: training snapshots end on 31 December 2024 and test snapshots cover 31 March–30 June 2025. Customer and snapshot identifiers were excluded from predictors. Historical recency, order counts, sales value, basket value, tenure and home-shop category were used.

Constant-prevalence, logistic-regression, random-forest and HistGradientBoosting candidates were compared. The final training fit uses all supplied TRAIN rows; the final TRAIN cohort is used for candidate and threshold selection. Test metrics include average precision, ROC-AUC, Brier score, thresholded precision/recall/F1, top-decile lift and a customer-cluster bootstrap. The target and predictors remain conditional on the supplied snapshot and order package.

4.11 RQ3 Scenario Policy Pipeline

The scenario runner compares the supplied baseline and trend-policy outcomes on the same 2,500 rows, one cutoff date, 32 shops and 477 SKUs. It verifies the supplied scenario cost-gap arithmetic and joins product purchase costs to check order-budget feasibility. Paired bootstrap intervals resample scenario rows; they describe variation across this scenario file, not sampling uncertainty for participating businesses. Future demand is the generated scenario-oracle field, so the runner does not claim a real-owner policy comparison or realised savings.

4.12 Multi-Date Validation and Owner-Impact Runner

The supplemental runner `run_multidate_business_impact.py` applies the same point-in-time rule to all three operational tasks. It writes one aggregate CSV per research question, a run manifest with input checksums and parameters, and three report figures. RQ1 evaluates six seven-day cutoffs; RQ2 evaluates six rolling customer snapshots; and RQ3 evaluates eight historical 30-day cutoffs using observed sales after the cutoff. The runner does not copy raw customer or transaction identifiers into the committed outputs. The RQ3 cost function is deliberately exposed as a parameterised assumption so that an owner can replace the 2% holding factor and margin proxy with documented business costs during a later replay.

4.13 RQ4 Explanation-Case Audit

The case audit re-evaluates whether stock plus pending orders crosses the reorder point, whether the scenario budget can buy a case pack, and whether the suggested quantity is a pack multiple within the stated budget. It also checks whether explanation text mentions stock, reorder point, pack multiple and budget. This is a mechanical consistency check. The supplied `human_rating` field has no participant, consent, task, timestamp or response provenance, so it is excluded from all usability calculations. The runner also produces a two-panel evidence figure: mechanical pass rates on the left and prepared-case versus validated-participant counts on the right.

Chapter 5

Experimental Results

5.1 Evidence Boundary and Panel Integrity

The supplied dataset manifest claims real SME business data, but its generation-seed and "modeled world" fields leave source provenance unverified. The shop–category panel contains 561,152 observations, 512 unique series and 1,096 daily observations per series. It totals 1,580,361 units; 37.5962% of rows are explicit zero-sales observations. The audit found no duplicate shop–category–date keys, missing required values, zero-flag mismatches or missing dates within the complete panel. Transaction and inventory reconciliation findings are reported in Chapter 4.

5.2 Dataset-to-Decision Analysis and Integrated Results

The dataset is not one homogeneous modelling table. Each research question has a different analytical grain, target and decision loss. Treating every row as an interchangeable training example would hide stock censoring, class imbalance and the difference between a prepared rule case and a real user response. The experiment design therefore follows the data structure:

Table 5.1: How the supplied data structure determines the evaluation design.

RQ	**Data evidence**	**Design problem**	**Evaluation response**	**Decision output**
RQ1	561,152 shop–category–day rows; 512 series; 37.60% zero-sales rows; many lumpy series.	Daily SKU sales are sparse and can be censored by unavailable stock.	Compare a seven-day shop–category forecast across six dates, then keep the one-day SKU exposure diagnostic separate.	Planning quantity with actual-versus-predicted demand and date-specific error.
RQ2	72,146 customer snapshots; 10,613 customers; 90-day inactivity labels.	Inactivity is imbalanced and repeated customers create temporal dependence.	Use chronological rolling fits, PR-AUC/Brier score and highest-risk-decile lift rather than accuracy alone.	A small, ranked retention list for limited owner resources.
RQ3	2,500 policy scenarios plus eight observed-sales historical cutoffs; reconstructed inventory is stock-rich.	Scenario cost is not realised profit, and high stock can conceal service differences.	Separate scenario results from observed-sales backtest and decompose purchase, holding and lost-margin cost.	Policy comparison with an explicit cash/service trade-off.
RQ4	1,800 prepared explanation cases; no participant linkage or consent record.	A prefilled rating field cannot establish usability or human performance.	Audit rule, constraint, pack and text consistency, then report participant evidence separately as zero.	Owner-readable explanation mechanism with a clear evidence boundary.

These checks directly determine what a defensible result can mean. The large zero-sales share makes percentage error unstable at the individual series level, so RQ1 reports a seven-day aggregate view alongside the exposure-aware SKU diagnostic. Repeated customer snapshots make random splitting unsafe, so RQ2 uses chronological rolling validation and evaluates the limited top-risk list. The reconstructed inventory is stock-rich, so RQ3 must report cost and lost units separately rather than calling a zero-stockout result a guaranteed profit. Finally, the prepared RQ4 cases can establish rule consistency but not human usability; that is why the participant evidence is shown as absent instead of being inferred from a prefilled rating field.

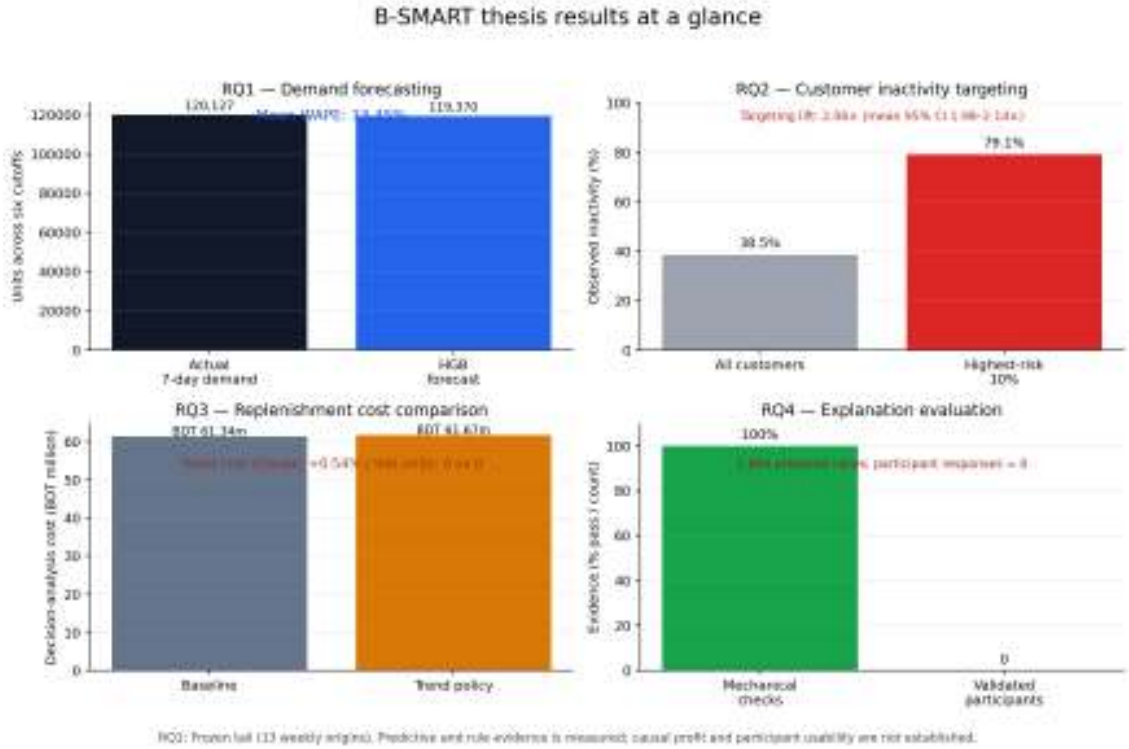


FIGURE 5.1: The complete thesis result in one view. RQ1 compares realised and forecast units, RQ2 shows the targeting lift, RQ3 shows the policy cost trade-off, and RQ4 distinguishes mechanical rule evidence from the absent participant study.

The novelty is therefore methodological and decision-oriented rather than a claim of inventing a new classifier. The framework combines (i) exposure-aware target construction for sparse SME demand, (ii) date-aware validation across multiple operational cutoffs, (iii) customer ranking evaluated at the budget- relevant top decile, (iv) replenishment evaluation that links forecasts to case-pack, stock and cost constraints, and (v) an explanation audit that does not confuse rule consistency with human usability. This design connects machine-learning outputs to owner decisions while keeping unsupported causal or profit claims outside the results.

For reader-facing communication, the following result figures deliberately show one decision at a time: realised demand versus the forecast, the practical customer targeting lift, and the historical replenishment cost trade-off. The full date-level metrics and comparator panels remain in the accompanying CSV files and diagnostic figures, but the primary figures answer the business question without requiring the reader to decode several axes at once.

5.3 Demand-Regime Distribution

Regimes are calculated using data available through 30 June 2025, before the first development origin. The ADI and CV^2 thresholds are 1.32 and 0.49. The panel contains 342 lumpy series, 153 erratic series, 13 smooth series and only 4 intermittent series.

TABLE 5.2: Historical demand-regime distribution across 512 shop–category series.

Regime	Series	Share	Mean zero-sales ratio
Lumpy	342	66.80%	47.62%
Erratic	153	29.88%	15.65%
Smooth	13	2.54%	3.64%
Intermittent	4	0.78%	83.94%
Total	512	100.00%	37.60%

Four intermittent series are not enough independent support for regime-specific automatic model switching. Repeated forecasts of the same four series do not create additional independent businesses.

5.4 Multi-Model Forecasting Performance

The fixed-origin experiment uses 1 July, 1 August and 1 September 2025. Each of the 512 series is forecast at each origin, giving 1,536 series-origin observations per horizon. The six candidates are LightGBM Tweedie, CatBoost Tweedie, a raw two-stage CatBoost hurdle, weekly seasonal-naive, Croston–SBA and TSB. All models use the same target definition and historical information at each origin.

TABLE 5.3: Pooled development WAPE by forecast horizon and candidate model.

Horizon	LightGBM WAPE	CatBoost WAPE	Hurdle WAPE	Croston–SBA WAPE	TSB WAPE	Seasonal naive
1 day	79.80%	79.98%	**79.23%**	84.01%	84.78%	92.21%
7 days	**33.74%**	34.20%	35.26%	37.74%	38.62%	40.77%
14 days	**24.11%**	24.78%	24.54%	30.27%	32.10%	33.97%
28 days	**18.57%**	19.28%	18.84%	25.62%	27.38%	29.16%

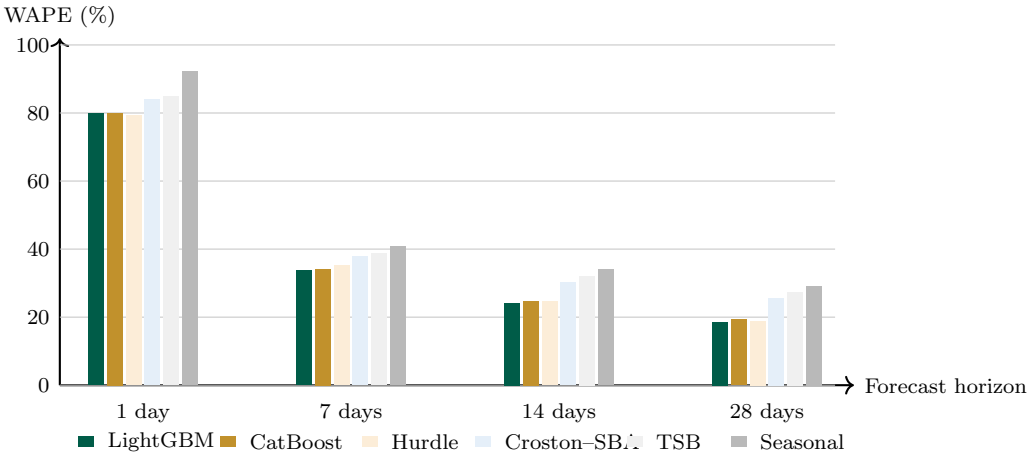


FIGURE 5.2: Measured development WAPE for the six candidates at four direct cumulative horizons. The one-day task remains difficult; lower WAPE at longer horizons reflects a different cumulative target and must not be interpreted as equivalent daily accuracy.

LightGBM has the lowest pooled WAPE at seven, fourteen and twenty-eight days. The raw hurdle candidate has the lowest one-day WAPE, but its one-day aggregate bias is −28.18%, compared with −21.69% for LightGBM. The hurdle probabilities are uncalibrated. Its lower one-day WAPE therefore does not justify promotion without calibration and a separate cost-sensitive analysis of under-forecasting.

Table 5.4: Forecast bias and horizon-scaled weekly MASE for the principal candidates.

Horizon	**LightGBM bias**	**CatBoost bias**	**Hurdle bias**	**LightGBM MASE**
1 day	−21.69%	−21.24%	−28.18%	0.793
7 days	−6.59%	−7.64%	−12.80%	0.326
14 days	−2.42%	−2.86%	−2.86%	0.226
28 days	−2.10%	−2.84%	−1.01%	0.175

Horizon-scaled weekly MASE divides absolute forecast error by the series-specific mean absolute lag-7 difference multiplied by the forecast horizon. It is a scale-normalised diagnostic, not a direct business-cost measure. The one-day WAPE remains high; longer-horizon results must not be interpreted as evidence of equally accurate daily replenishment forecasts.

5.5 Seven-Day Stability and Shop-Level Heterogeneity

LightGBM has the lowest seven-day pooled WAPE at each development origin: 32.52% on 1 July, 33.86% on 1 August and 34.92% on 1 September. CatBoost records 32.71%, 34.26% and 35.72%; the raw hurdle records 33.81%, 35.54% and 36.52%.

Table 5.5: Seven-day WAPE by development origin.

Origin	**LightGBM**	**CatBoost**	**Hurdle**	**Croston–SBA**	**TSB**
1 July 2025	**32.52%**	32.71%	33.81%	37.91%	39.82%
1 August 2025	**33.86%**	34.26%	35.54%	36.14%	35.82%
1 September 2025	**34.92%**	35.72%	36.52%	39.21%	40.24%

Across shops, LightGBM has lower seven-day WAPE than CatBoost in 22 of 32 shops; across categories it wins in 14 of 16. Median shop-level WAPE is 35.89% for LightGBM and 36.79% for CatBoost. The model is not uniformly best for every shop, so per-shop monitoring and abstention remain necessary.

Table 5.6: Cluster-bootstrap seven-day WAPE differences.

Challenger	**Median difference**	**95% interval**	**Interpretation**
CatBoost Tweedie	−0.45 pp	[−0.85, −0.08] pp	LightGBM lower WAPE
Raw CatBoost hurdle	−1.51 pp	[−2.12, −0.93] pp	LightGBM lower WAPE
Weekly seasonal-naive	−7.01 pp	[−8.55, −5.59] pp	LightGBM lower WAPE
Croston–SBA	−3.97 pp	[−5.04, −2.94] pp	LightGBM lower WAPE
TSB	−4.90 pp	[−6.11, −3.72] pp	LightGBM lower WAPE

The intervals resample 512 shop–category series with 2,000 bootstrap replicates. They quantify variation across the development series under the fixed-origin protocol; they are not uncertainty intervals for future real-world SME performance and do not replace an untouched test.

5.6 Historical Transaction-Feature Ablation

The transaction-derived feature experiment evaluates nine monthly development origins from January to September 2025, with 512 series at each origin and 4,608 series-origin forecasts per model. The baseline uses the historical demand features only. The extended candidate adds historical price, discount, order and customer-day summaries, all computed strictly before the forecast origin.

Table 5.7: Nine-origin seven-day LightGBM transaction-feature ablation.

Feature set	**Pooled WAPE**	**Mean origin WAPE**	**Bias**	**MAE**	**RMSE**
Historical demand only	**33.7313%**	**34.1557%**	-1.35%	**7.124**	**9.824**
Historical price/activity	33.8358%	34.2685%	-1.28%	7.146	9.866

The extended features improve WAPE at five origins and worsen it at four, but the pooled WAPE is 0.1045 percentage points higher and mean origin MAE is also higher. Aggregate under-forecast is marginally smaller (-1.28% versus -1.35%), showing that bias and absolute error do not always move together. The evidence does not support adding these transaction features to the current selected model. They are retained as a reproducible ablation, not adopted as a performance improvement. Future covariates should be added only when point-in-time availability is established and they improve pooled and origin-stability metrics.

5.7 Performance by Demand Regime

At seven days, CatBoost has slightly lower WAPE than LightGBM in the erratic group (28.50% versus 28.67%), while LightGBM performs better in the lumpy and smooth groups. Croston–SBA has lower WAPE in the intermittent group (77.02% versus 106.30% for LightGBM), but that subset contains only four series, twelve series-origin forecasts and 25 actual units. The raw hurdle has WAPE 83.18% in the same subset but under-forecasts total demand by 74.62%. None of these sparse subgroup results justifies automatic model routing.

Table 5.8: Seven-day WAPE by historical demand regime.

Regime	Cases	LightGBM	CatBoost	Hurdle	Croston–SBA
Erratic	459	28.67%	**28.50%**	28.57%	32.38%
Lumpy	1,026	**42.06%**	42.85%	45.60%	47.09%
Smooth	39	**19.89%**	22.03%	20.60%	20.69%
Intermittent	12	106.30%	121.62%	83.18%	**77.02%**

Table 5.9: Seven-day WAPE for the context challenger and baseline at each of nine monthly origins.

Origin	Baseline WAPE	Context WAPE	Change (pp)	Lower error
1 January 2025	34.34%	33.80%	-0.54	Context
1 February 2025	36.06%	35.54%	-0.52	Context
1 March 2025	31.07%	32.88%	+1.81	Baseline
1 April 2025	29.23%	28.36%	-0.87	Context
1 May 2025	31.90%	31.50%	-0.40	Context
1 June 2025	32.34%	33.50%	+1.17	Baseline
1 July 2025	40.83%	32.34%	-8.48	Context
1 August 2025	34.01%	33.71%	-0.29	Context
1 September 2025	34.82%	34.92%	+0.09	Baseline

The context features do not win at every origin. The largest improvement occurs in July, while March, June and September favour the baseline. This variation is important: a pooled reduction must not be presented as uniform improvement across time. It supports retaining the baseline as a fallback and monitoring the context model by origin and business.

5.8 Context-Aware Model Results

Table 5.10 compares the historical-feature baseline with the context-aware challenger on identical development origins. At seven days, pooled WAPE decreases from 33.47% to 32.66%, an absolute reduction of 0.81 percentage points (2.41% relative). The challenger

wins at six of nine monthly origins. A paired 2,000-replicate shop–category cluster bootstrap estimates the context-minus-baseline WAPE difference at -0.81 percentage points, with a 95% interval of [-1.09, -0.52] percentage points.

Table 5.10: Context-aware LightGBM versus the same-model historical-feature baseline.

Horizon	Baseline WAPE	Context WAPE	Absolute change	Context bias
1 day	82.73%	79.74%	-2.99 pp	-16.26%
7 days	33.47%	32.66%	-0.81 pp	-6.05%
14 days	28.48%	24.93%	-3.56 pp	+5.82%
28 days	19.35%	18.46%	-0.89 pp	+2.67%

The seven-day result is the strongest comparison because it uses nine origins; other horizons use three origins each. The context model's seven-day bias is -6.05%, compared with -2.02% for the baseline. A past-origin-only multiplicative adjustment reduced bias to -1.61% but increased WAPE to 32.90%. It is an optional cost-sensitive trade-off, not silently applied to the selected forecast.

Prequential scaled-residual intervals, calibrated only from earlier validation origins, achieved mean empirical coverage of 79.43% for nominal 80% intervals and 89.55% for nominal 90% intervals across the evaluated origins. These are development diagnostics, not guarantees of future coverage.

Bottom-up aggregation of the same seven-day leaf forecasts is coherent. WAPE is 32.66% at shop–category level, 14.95% at shop total, 10.98% at category total and 7.85% at network total. Lower aggregate errors partly reflect cancellation and do not establish equivalent leaf-level replenishment accuracy. The context model is the leading development challenger; production selection requires a genuinely untouched future period or held-out businesses and explicit under-forecast cost evaluation.

Table 5.11: Seven-day pooled error and bias diagnostics for the paired context experiment.

Metric	Baseline	Context model
Pooled WAPE	33.47%	32.66%
Mean origin WAPE	33.84%	32.95%
MAE (units per series-origin)	7.069	6.898
RMSE (units per series-origin)	10.008	9.792
Aggregate signed bias	-2.02%	-6.05%

The paired comparison uses the same 512 shop–category series at all nine origins. The 2,000-replicate cluster bootstrap resamples complete series clusters, retaining each sampled series' repeated-origin structure. The median WAPE difference (context minus baseline) is -0.811 percentage points, with a 95% interval from -1.091 to -0.517 percentage

points. This interval quantifies variation across the development series; it is not a prediction interval for a future business or a substitute for the untouched final test.

Table 5.12: Prequential interval coverage for the seven-day context forecast.

Origin	**Calibration origins**	**80% coverage**	**90% coverage**	**Interpretation**
1 April 2025	3	75.78%	89.06%	80% under-covered
1 May 2025	4	84.77%	92.77%	Both above nominal
1 June 2025	5	71.29%	81.25%	Both under-covered
1 July 2025	6	82.81%	92.58%	Both above nominal
1 August 2025	7	82.62%	91.99%	Both above nominal
1 September 2025	8	79.30%	89.65%	Close to nominal

The interval method uses scaled absolute residuals from earlier origins only. Mean coverage across these six evaluated origins is 79.43% for the nominal 80% interval and 89.55% for the nominal 90% interval. Coverage nevertheless varies materially by month. The interval estimates are therefore useful diagnostics for uncertainty communication but are not yet sufficient to guarantee a fixed coverage level in deployment.

5.9 Multi-Date Out-of-Time Demand Validation

The fixed-origin results above are complemented by a second date-based evaluation designed to answer a practical owner question: when the forecast is made at different points in the year, how close is it to the demand that actually follows? Six month-end cutoffs (30 April–30 September 2025) were used. At each cutoff, the model was trained only on earlier seven-day origins within a trailing 365-day window and then predicted the next seven days for all 512 shop–category series. The two comparators were a weekly seasonal naive forecast and the previous-28-day mean. Thus, the date of the decision is always earlier than the measured outcome.

For a series j at cutoff t, the two simple forecasts are

$$\hat{y}_{j,t}^{(7)} = \sum_{h=1}^{7} y_{j,t-h}, \qquad \hat{y}_{j,t}^{(28)} = 7\left(\frac{1}{28}\sum_{h=1}^{28} y_{j,t-h}\right).$$

The HistGradientBoosting model uses lagged demand, rolling means and dispersion, series history and calendar encodings. It is fitted with a Poisson loss, which penalises errors on a non-negative count target while preserving the chronological split. The reported network totals are sums of the 512 leaf predictions; WAPE is $\sum_j |\hat{y}_j - y_j| / \sum_j |y_j|$.

TABLE 5.13: Six-date seven-day demand validation at the shop–category level.

Cutoff	Actual units	HGB forecast	HGB WAPE	28-day WAPE	Weekly WAPE	HGB signed bias
30 Apr 2025	9,978	10,069	31.38%	39.27%	43.38%	+91
31 May 2025	12,659	11,314	29.18%	36.11%	43.79%	-1,345
30 Jun 2025	10,010	9,565	32.36%	46.11%	46.75%	-445
31 Jul 2025	9,793	9,505	33.80%	36.70%	43.95%	-288
31 Aug 2025	9,417	9,214	34.50%	39.32%	48.74%	-203
30 Sep 2025	9,125	9,314	34.20%	38.47%	49.74%	+189

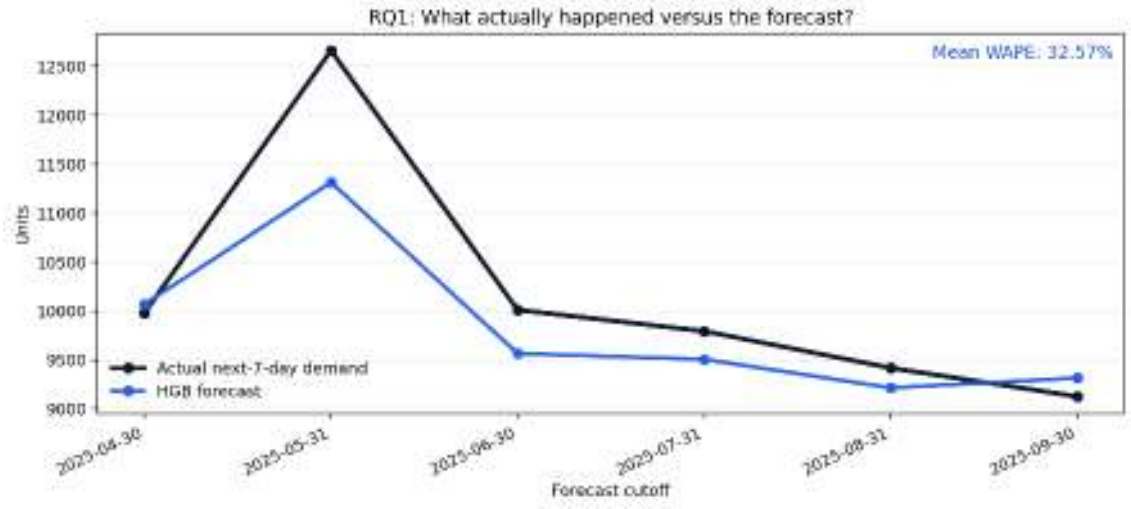


FIGURE 5.3: Reader-facing RQ1 result: realised next-seven-day demand versus the HistGradientBoosting forecast at six chronological cutoffs. The mean WAPE is shown directly on the graph; the date-level comparator table retains the full diagnostic detail.

Across the six cutoffs, HistGradientBoosting has mean WAPE 32.57%, compared with 39.33% for the 28-day mean and 46.06% for the weekly naive rule. Its aggregate signed bias is -3.29% over 60,982 actual units. The model is therefore materially more useful for this seven-day, shop–category decision than the two simple comparators in this multi-date test, although its WAPE still ranges from 28.86% to 34.23%. This result must not be merged with the separate one-day SKU exposure diagnostic, where the target grain, horizon and stock-eligibility rule are different and the 28-day mean remained the best WAPE baseline. The multi-date result is a stronger operational diagnostic, not a claim that every SKU forecast is accurate.

TABLE 5.14: Frozen-tail RQ1 stress check after freezing the context-aware model.

Model	Origins	Actual units	Forecast units	WAPE	Bias
Base LightGBM	13	120,127	118,831.0	33.58%	-1.08%
Context-aware LightGBM	13	120,127	119,369.6	**33.45%**	-0.63%

The model was fitted once through 30 September 2025 and then scored at 13 weekly origins from 1 October to 24 December 2025. All rolling and cross-series features stop before each origin. Context features reduce WAPE by 0.13 percentage points and reduce aggregate under-forecast bias, but this is a pre-inspected Q4 tail stress check rather than a blind final holdout. The graphs below show both the unit-level outcome and the error split by demand regime.

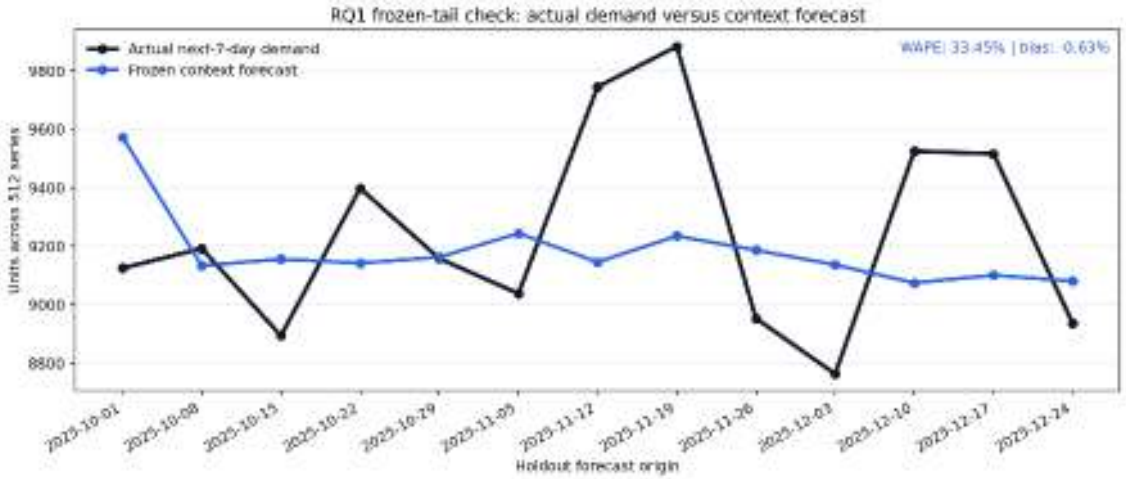


FIGURE 5.4: RQ1 frozen-tail stress check: actual versus frozen context-aware forecast across 13 weekly origins.

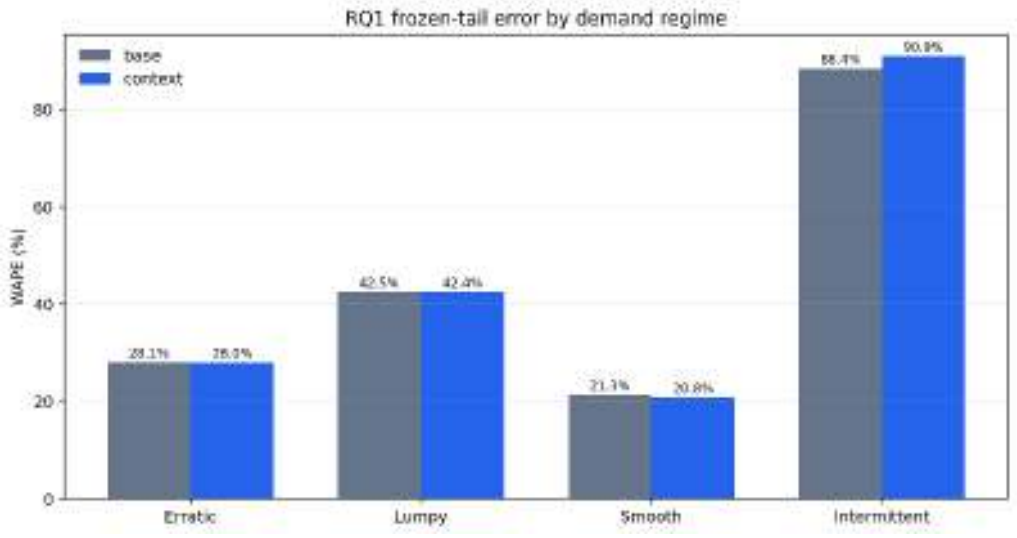


FIGURE 5.5: RQ1 frozen-tail WAPE by demand regime. Regimes use training-period history only.

Table 5.15: Bottom-up seven-day forecast accuracy at different aggregation levels.

Aggregation level	Forecast cases	WAPE	Signed bias
Shop–category	4,608	32.66%	-6.05%
Shop total	288	14.95%	-6.05%
Category total	144	10.98%	-6.05%
Network total	9	7.85%	-6.05%

The hierarchy is coherent because shop, category and network totals are sums of the same shop–category leaf forecasts rather than separately fitted aggregate models. Error decreases as values are aggregated partly because positive and negative series errors can offset one another. Consequently, network-level accuracy cannot replace shop–category accuracy when the decision is a local reorder.

5.10 Model and Solution Design Decision

The current forecasting design retains the original pooled LightGBM Tweedie as a benchmark and advances the context-aware LightGBM as the leading development challenger. It uses the same model family and settings while adding historical shop, category and network aggregates. The nine-origin seven-day comparison shows lower pooled WAPE and a paired cluster-bootstrap interval favouring the context features. However, its greater under-forecast bias means the final decision must consider the business cost of shortages, not WAPE alone.

The current selection is therefore conditional: use the context-aware candidate for further validation, retain the baseline as fallback, and do not promote a production model until an untouched future period or held-out businesses have been evaluated. CatBoost remains a challenger; the hurdle model remains experimental until occurrence probabilities are calibrated and bias is controlled. Croston–SBA and TSB remain intermittent-demand comparators, but sparse regime support does not justify automatic routing.

The separate SKU diagnostic assigns zeros only to reconstructed positive-stock days and quarantines all 253 histories with negative balances. It is an exposure-conditioned observed-sales evaluation, not latent-demand estimation. The RQ3 scenario analysis and the new historical observed-sales backtest are reported below. They provide decision-analysis evidence across dates, but do not establish audited owner-policy profit. The RQ4 explanation-case audit also remains a mechanical consistency check without participant usability evidence.

5.11 RQ1 Exposure-Conditioned SKU Diagnostic

Inventory movements were reconstructed chronologically. The panel retained 13,545,871 exposed shop–SKU–days after quarantining 253 histories with negative balances. It contains 685,827 positive days and 12,860,044 zeros assigned only when reconstructed stock remained positive throughout the day. The diagnostic training window ends on 30 September 2025. Sequential one-day predictions cover 1 October–31 December 2025, with prior observed outcomes available to subsequent daily predictions. The Q4 period had been inspected earlier in project work and is not an untouched final holdout.

The evaluation includes 1,130,148 exposed shop–SKU–days from 13,019 series, with 119,161 observed units and a 95.34% zero-target share. The HistGradientBoosting model used all eligible positive training days and a weighted sample of 500,000 exposed-zero days. The comparators are a weekly same-day seasonal-naive rule and the mean over the previous 28 exposed days.

Table 5.16: Exposure-conditioned one-day SKU results on the previously inspected Q4 period.

Model	WAPE	MAE	RMSE	Bias
Weekly seasonal naive	190.28%	0.201	0.853	+1.19%
28-day exposed-sales mean	**183.65%**	**0.194**	0.616	+0.82%
HistGradientBoosting	184.66%	0.195	**0.604**	+2.06%

The 28-day mean has the lowest WAPE; HistGradientBoosting is 1.01 percentage points worse on that metric, although it has a slightly lower RMSE. A 2,000-replicate shop-cluster bootstrap for the HGB-minus-28-day-mean WAPE difference is +1.01 percentage points, with a descriptive 95% interval of -0.21 to +2.25 points. The interval includes zero and is based on only 32 shops. HGB improves on weekly seasonal naive, but does not outperform the strongest simple baseline on WAPE. The result does not support promoting the machine-learning candidate for this task.

This diagnostic evaluates observed sales conditional on reconstructed full-day stock exposure. It does not recover demand during stock-outs. Ledger-based exposure is not independently checked against physical stock, and this reporting period is not blind; RQ1 therefore remains incomplete.

5.12 RQ2 Customer Inactivity Prediction

The 90-day inactivity labels were reconstructed from the supplied sales orders. There were no missing customer histories and no label mismatches among 72,146 snapshots. The temporal TRAIN split contains 50,982 rows through 31 December 2024; the TEST

split contains 21,164 rows from 31 March–30 June 2025. The test inactivity rate is 41.18%, compared with 36.39% in TRAIN. Repeated customers appear on both sides of this temporal split, so uncertainty intervals resample customer clusters; customer IDs were not used as model features.

Models were selected on the final TRAIN snapshot cohort (31 December 2024) by average precision, then refitted on all TRAIN rows. Each candidate's decision threshold was selected on that validation cohort by F1. The HistGradientBoosting validation average precision was 0.717, the highest of the three fitted candidates.

Table 5.17: Customer-inactivity model comparison on the supplied temporal TEST split.

Candidate	**PR-AUC**	**ROC-AUC**	**Brier**	**F1**
Constant TRAIN prevalence	0.412	0.500	0.245	0.000
Logistic regression	0.739	0.823	0.178	0.710
HistGradientBoosting	**0.746**	**0.830**	**0.168**	0.716
Random forest	0.732	0.825	0.170	**0.718**

For the selected HistGradientBoosting model, the validation-selected threshold is 0.222. TEST precision is 0.588, recall is 0.913, and precision in the highest-scored 10% is 0.845 (2.05 times the test base rate). A 300-replicate customer-cluster bootstrap gives a 95% interval of 0.736–0.758 for PR-AUC and 0.326–0.345 for its paired improvement over the constant-prevalence baseline. This supports ranking performance on the supplied snapshots, not the causal effect of a retention action. Source provenance remains unverified.

5.13 Multi-Date Customer-Risk Stability and Targeting Lift

The supplied temporal split gives one useful test, but a retention system also needs to remain useful as the snapshot date changes. A rolling evaluation was therefore run at six snapshot dates from 31 March 2024 to 30 June 2025. For each date, all earlier snapshots were used for fitting and only the customers at the current date were scored. No later snapshot or future label was used in the feature matrix. The model output is $p_i = P(y_i = 1 \mid x_i)$, the estimated probability that customer i will become inactive in the following 90 days. Average precision (AP) measures ranking quality under class imbalance, while the Brier score measures the mean squared probability error.

TABLE 5.18: Rolling customer-inactivity validation and the highest-risk decile.

Snapshot	Actual inactive	Mean predicted probability	AP	Top-10% observed inactive	Brier
31 Mar 2024	32.15%	28.47%	0.619	69.91%	0.165
30 Jun 2024	37.79%	30.17%	0.684	77.90%	0.178
30 Sep 2024	39.46%	35.43%	0.705	79.98%	0.172
31 Dec 2024	39.51%	27.97%	0.716	79.60%	0.185
31 Mar 2025	39.01%	35.61%	0.722	81.16%	0.166
30 Jun 2025	43.34%	38.01%	0.767	86.06%	0.168



FIGURE 5.6: Reader-facing RQ2 result: observed inactivity for all customers versus the highest-risk 10% selected by the model. The graph reports the average targeting lift directly; chronological metrics and calibration diagnostics are retained in the table.

The top-risk decile has an observed inactivity rate of 69.91–86.06%, whereas the corresponding full-population rate is 32.15–43.34%. This is approximately a 1.99–2.17-fold targeting lift, with a mean lift of 2.06x; date-level bootstrap intervals average 1.98–2.14x. A small SME can therefore direct a limited retention budget to a much smaller customer list. The mean predicted probability is below the observed rate at all six dates, with a gap of 3.40 to 11.54 percentage points. The model is therefore useful as a ranking tool, but its probabilities require calibration before they are interpreted as literal customer-level percentages or used to set an automatic contact threshold. The experiment demonstrates predictive targeting value, not that an SMS, discount or call caused the subsequent order.

5.14 RQ3 Replenishment Scenario Comparison

The supplied file contains 2,500 scenarios at the 31 July 2025 cutoff, covering 32 shops and 477 SKUs. Its future-demand field is explicitly oracle future-demand assumptions. The

baseline and trend policy were compared on identical scenarios; product costs confirmed zero order-budget breaches for either policy.

TABLE 5.19: Paired replenishment outcomes under the supplied scenario assumptions.

Scenario outcome	**Baseline**	**Trend policy**	**Change**
Total scenario cost (BDT)	2,227,690.07	2,208,289.47	-19,400.60 (-0.87%)
Lost units in scenarios	45	100	+55
Mean average stock (units)	157.13	155.03	-2.10
Order-budget breaches	0	0	0

The paired scenario bootstrap estimates a mean trend-minus-baseline cost difference of -BDT 7.76 per scenario (95% interval -9.26 to -6.30) and 0.022 additional lost units per scenario (95% interval 0.012 to 0.034). Under these assumptions, lower scenario cost accompanies more lost units; the trend policy is not superior on both outcomes. The figures are not realised savings and do not compare against a participating owner's documented policy.

5.15 Historical Multi-Date Replenishment Backtest and Owner Impact

The paired scenario comparison is useful for controlled policy testing, but it does not show what would have happened after a real historical cutoff. A second backtest therefore uses observed sales after eight month-end cutoffs from 28 February to 30 September 2025. For every shop–SKU pair, the next 30 days of observed sales are kept as the outcome. Stock at the cutoff is reconstructed from inventory movements, while demand estimates use only the preceding 30 and 90 days.

The baseline and trend forecasts are defined as

$$\hat{D}_{30}^{\text{base}} = 30(D_{90}/90), \qquad \hat{D}_{30}^{\text{trend}} = 30\left[0.6(D_{30}/30) + 0.4(D_{90}/90)\right].$$

For forecast $\hat{D}$, stock I_t, reorder point r and case pack k, the proposed order is

$$Q = k\left\lceil \frac{\max(0, \hat{D} + r - I_t)}{k} \right\rceil.$$

To make the owner trade-off explicit, the decision-analysis cost is

$$C = Qc + 0.02\,\bar{I}c + L\max(p - c, 0),$$

where c is purchase cost, $\bar{I}$ is average post-order stock, L is lost units and p is the catalogue retail price. The 2% holding factor is an assumption for comparing policies; it is not an audited accounting rate.

TABLE 5.20: Observed-sales replenishment backtest across eight historical cutoffs.

Cutoff	Actual next-30d units	Baseline forecast	Trend forecast	Trend cost change	Lost units (both)	Pairs
28 Feb 2025	50,747	39,877	40,006	+0.21%	0	12,197
31 Mar 2025	51,113	43,831	48,432	+1.85%	0	12,300
30 Apr 2025	41,546	47,574	49,698	+0.84%	0	12,289
31 May 2025	53,803	47,974	44,083	-0.44%	0	12,422
30 Jun 2025	40,632	48,529	51,693	+1.57%	0	12,377
31 Jul 2025	42,010	45,099	42,498	-0.00%	0	12,305
31 Aug 2025	40,160	45,425	42,912	-0.13%	0	12,311
30 Sep 2025	39,222	40,916	40,462	+0.12%	0	12,183

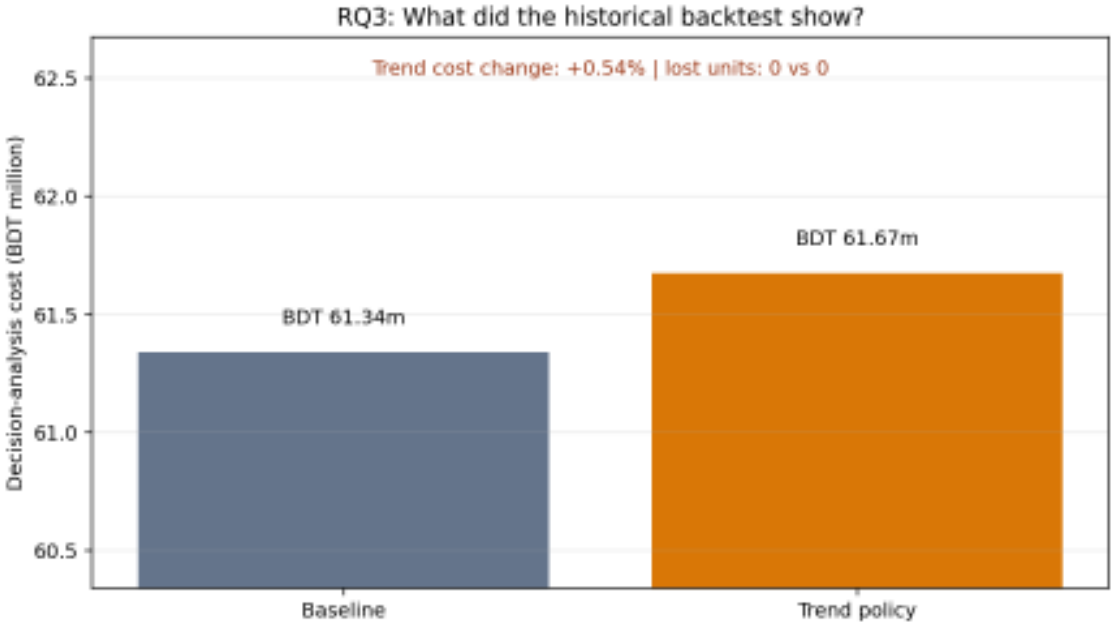


FIGURE 5.7: Reader-facing RQ3 result: total historical decision-analysis cost for the baseline and trend policies. The trend policy is 0.54% more expensive in this stock-rich sample, while both policies record zero lost units.

Across all eight cutoffs, baseline cost is BDT 61.341 million and trend-policy cost is BDT 61.675 million, a weighted increase of 0.54%. The trend policy is cheaper at only three of eight cutoffs and more expensive at five. Both policies record zero lost units because reconstructed stock is high relative to the observed 30-day demand; this is evidence about the selected historical sample, not proof that either policy prevents stock-outs in a different SME. The low-stock quartile file repeats the comparison for the least-stocked pairs. A holding-cost sensitivity check at 0%, 2%, 5% and 10% changes the absolute cost level but not the service result: lost units remain zero for both policies. A service-cost Pareto frontier is therefore not identifiable from this stock-rich sample. The practical contribution is decision visibility: an owner can see when trend adjustment changes procurement

and holding cost, while avoiding a false claim of guaranteed profit. A real owner-policy replay with validated purchase, holding and lost-sales costs is still required before financial deployment.

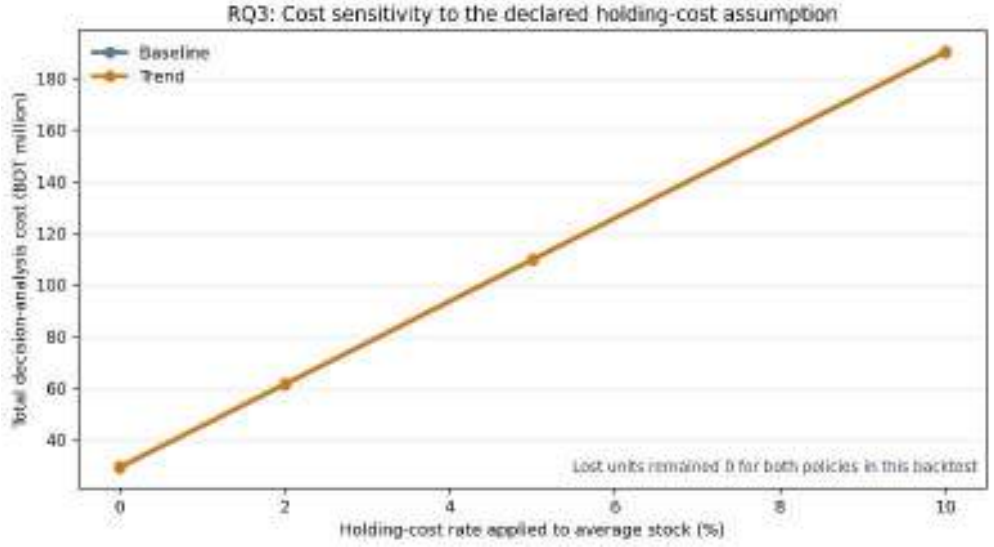


FIGURE 5.8: RQ3 sensitivity to the declared holding-cost rate. The two policies remain close, while the observed-sales sample provides no lost-unit separation.

5.16 Business-Owner Decision Value

The experiments translate the modelling outputs into three decisions that a small business owner can understand and audit. Demand forecasting estimates how many units may be needed soon; inactivity prediction ranks which existing customers should receive limited retention attention; and replenishment optimisation compares the cash, stock and service consequences of an order rule. The evidence is useful because each decision is shown against an observed outcome at several dates, rather than represented by one overall accuracy number.

Table 5.21: How the research can support an SME owner's decisions.

Owner question	**Model evidence**	**Operational use**	**Evidence boundary**
How much should I prepare for next week?	RQ1 leakage-safe HGB mean WAPE 32.57% across six dates, plus a frozen-tail context check at 33.45%.	Set a shop–category planning quantity and review dates with larger bias or WAPE.	Not a guarantee for an unseen business; stock-out demand is not recovered.
Which customers need attention first?	RQ2 highest-risk 10% observed inactivity 69.91–86.06%, versus 32.15–43.34% overall.	Prioritise a small, affordable call, message or offer list instead of contacting everyone.	Ranking is useful, but probabilities need calibration and action impact was not randomised.
Which replenishment rule is cheaper?	RQ3 evaluates forecast, order quantity, stock, lost units and an explicit cost function at eight dates.	Compare the owner policy with baseline/trend alternatives and identify months needing review.	Costs are decision-analysis assumptions; both policies had zero lost units in this stock-rich sample.

The mathematical formulation makes the benefit inspectable: forecast error is measured against realised demand, customer ranking is measured against the later inactivity label, and policy cost is decomposed into purchase, holding and lost-margin terms. This supports an SME owner as a monitoring and prioritisation tool, not as an autonomous promise of profit. The next validation step is to log owner decisions and realised outcomes so that the same metrics can be recalculated with actual intervention, cost and service data.

5.17 RQ4 Explanation-Case Consistency Audit

All 1,800 supplied cases have the date 31 December 2025. The audit recomputed the action and constraint reason from inventory position, reorder point, case-pack size and scenario budget. All 1,800 matched. All suggested quantities were case-pack multiples within the stated scenario budget, and each explanation text mentioned stock, reorder point, pack multiple and budget. The result verifies rule and text consistency only; the case file identifies stock values as scenario data, not live inventory.

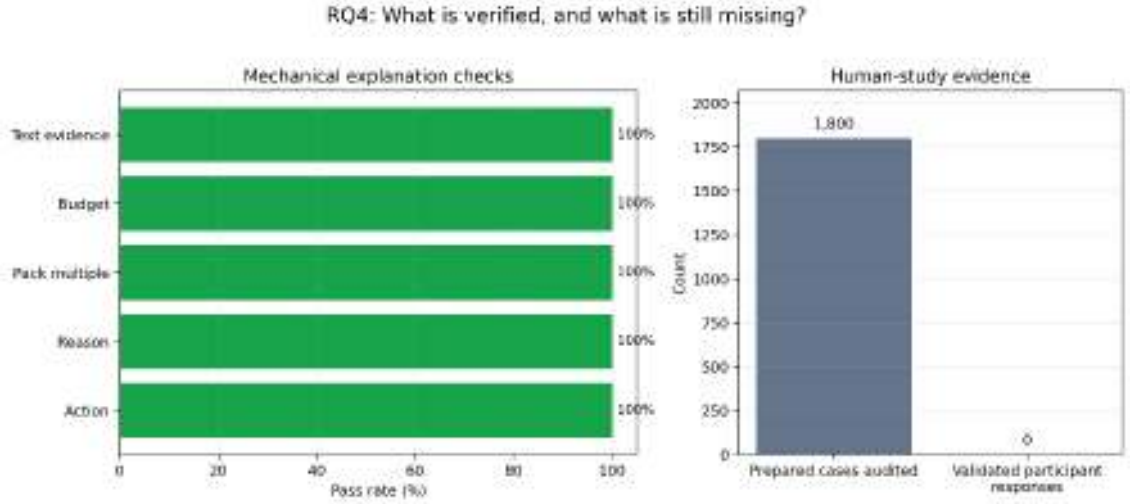


FIGURE 5.9: RQ4 evidence boundary. Every mechanical check passes for the 1,800 prepared cases, but the dataset contains zero validated participant responses. The graph makes clear why the current RQ4 result is an explanation-engine audit, not a usability study.

The supplied `human_rating` value is YES in all 1,800 rows, but there is no participant, consent, task, timestamp or response provenance. Validated participant responses are therefore zero, and no usability, confidence, task-correctness or decision-time metric can be reported. The real owner study for RQ4 remains incomplete.

5.18 Research-Question Evidence Status

TABLE 5.22: Current status of empirical evidence for the four research questions.

RQ	**Evidence completed**	**Remaining requirement**
RQ1	Multi-horizon development benchmark, six-date seven-day validation and exposure-conditioned SKU diagnostic	Untouched future test; independent stock and source validation; leaf-level cost evaluation
RQ2	Label reconstruction, supplied temporal split and six-date rolling ranking validation	Independent provenance and external business validation; probability calibration and no causal retention-effect claim
RQ3	Paired comparison on 2,500 scenario rows plus eight-date observed-sales backtest	Participating owner's actual policy and validated realised cost/outcome data
RQ4	Mechanical consistency audit of 1,800 prepared test cases	Consented participant study with measured task accuracy, completion time, confidence and usability

5.19 Threats to Validity and Summary

The results are measured on the researcher-supplied package, whose claimed business provenance remains unverified. RQ1 shop–category results are bounded to the supplied panel, selected development origins and cumulative forecast horizons. The exposure-conditioned SKU diagnostic uses Q4 2025, which had previously been inspected; neither that period nor the earlier development origins constitute an untouched final test. The new RQ3 historical backtest uses observed sales after each cutoff, but its cost terms remain assumptions and it does not replay an owner's documented policy. The separate RQ3 scenario comparison uses generated scenario values, and RQ4 uses prepared rule-test cases rather than participant responses. The earlier WAPE values of 83.98%, 84.04% and 84.56% are not compared with the current experiments because their target, aggregation, origins and metric denominators have not been reproduced on identical evaluation rows.

The evidence supports keeping context features as a shop–category development challenger while retaining the pooled baseline as fallback. The multi-date seven-day RQ1 validation is operationally stronger than the one-day SKU diagnostic, but it does not remove the need for an untouched test and stock validation. RQ2 supports high-risk customer ranking, not a causal retention effect. RQ3 shows how an owner can compare policies across dates, but not audited profit or universal stock-out prevention. RQ4 is only a rule consistency audit. Remaining empirical work includes independent source and stock validation, an untouched RQ1 test, calibrated customer probabilities, participating-owner policy replay, validated costs and a consented participant study. No universal model winner or untested business-impact claim is asserted.

Chapter 6

Conclusions

6.1 Research Summary

This thesis develops and evaluates an integrated SME analytics framework using the researcher-supplied Bangladesh data package. Its manifest claims real SME sales and people, but a generation seed and "modeled world" wording leave source provenance unverified. Structural checks found that sales lines join to orders, prepared SKU aggregates reproduce from transaction lines, category totals reproduce from SKU sales, and inventory sale movements reconcile to sales lines. These checks establish internal consistency, not real-world origin. The inventory audit also found negative-balance histories and positive-sale days with non-positive reconstructed stock; affected histories were quarantined from the exposure-conditioned SKU diagnostic.

For RQ1, the complete shop–category panel supports observed-sales forecasting. A context-aware LightGBM challenger adds historical shop, category and network aggregates to the baseline features. Across nine seven-day development origins, pooled WAPE improves from 33.47% to 32.66%, with a paired cluster-bootstrap interval supporting lower development error. However, under-forecast bias increases from -2.02% to -6.05%. The model is therefore a leading development challenger, not yet a final production selection. The one-, 14- and 28-day comparisons also show lower pooled WAPE, but use only three origins each.

An exposure-conditioned SKU diagnostic found that the 28-day exposed-sales mean had lower Q4 WAPE than HistGradientBoosting; the period had already been inspected and is not a blind holdout. A separate six-date seven-day validation recorded leakage-safe mean WAPE of 32.57% for HistGradientBoosting, with actual and predicted totals shown at every cutoff. For RQ2, temporal model comparison on the supplied snapshots produced a HistGradientBoosting PR-AUC of 0.746 after all 72,146 labels matched an independent order-history reconstruction; the rolling evaluation also placed 69.91–86.06% observed inactivity in the highest-risk 10%. For RQ3, the trend scenario had 0.87% lower cost but 55 more lost units than the supplied baseline, while the eight-date observed-sales backtest

showed a 0.54% weighted cost increase and zero lost units under both policies in a stock-rich sample. For RQ4, all prepared cases passed mechanical rule checks, but there are no validated participant responses. These findings are conditional on unverified source data, cost assumptions and the absence of causal intervention evidence. The contribution is an auditable, date-aware decision process; owner-policy benefit, measured usability, causal customer-action effects and transfer to other businesses remain unestablished.

The thesis novelty is the evidence design that connects these tasks without mixing their meanings: exposure-aware demand evaluation for sparse sales, rolling risk ranking for imbalanced inactivity, constraint- and cost-linked replenishment, and explanation consistency separated from human usability. This makes the framework useful for SME monitoring and prioritisation while keeping the claims proportionate to what the supplied data can actually prove.

Appendix A

RQ1 Data Audit and Target Eligibility

A.1 Analytical Grain and Field Definitions

The RQ1 source panel is defined at one row per shop, product category and calendar date. Its fields have the following roles:

Table A.1: Field dictionary for the complete RQ1 panel.

Field	Type	Meaning and validation
`shop_id`	Categorical key	Shop identifier; retained as a model feature and grouping key
`category`	Categorical key	Product category; retained as a model feature and grouping key
`date`	Date	Calendar date; parsed and checked for daily continuity
`units_sold_observed`	Non-negative numeric	Observed units sold in the shop-category-date cell; target quantity
`is_zero_observed_sales`	Binary indicator	Must equal one exactly when observed units are zero

The complete panel contains 512 shop–category series. Each series has 1,096 consecutive dates from 1 January 2023 to 31 December 2025. The resulting row count is therefore $512 \times 1{,}096 = 561{,}152$. The total observed quantity is 1,580,361 units. Explicit zero-sales rows number 210,972, giving a zero-sales share of 37.5962%.

A.2 Observed Zero, Missing Observation and Censored Demand

Three states must be kept separate:

1. **Observed zero:** a complete shop–category–date row exists, its quantity is zero and the zero indicator agrees. This is a valid observed-sales target for the current RQ1 task.

2. **Missing observation:** the expected row is absent. Its meaning is unknown until source completeness, operating calendar and assortment are checked. It is not automatically zero.

3. **Censored demand:** a product may have been wanted by customers but unavailable because of stock-out or other operating constraints. Recorded sales may then understate latent demand.

The complete shop–category panel passes the daily continuity and zero-indicator checks. The sparse SKU export does not provide a complete SKU-date panel. A missing SKU date can therefore represent inactive assortment, a closed shop, an incomplete export or an unavailable product. SKU-level latent-demand modelling is not enabled until product activity, shop operation and stock exposure are validated for each date.

The inventory reconstruction identified 269 negative-balance movement events across 253 shop–SKU combinations and 330 positive-sale days with non-positive reconstructed intraday stock. These events are retained as exceptions and excluded from any claim that the affected SKU-day was continuously in stock. The 2,088 theoretical shop–SKU combinations without an opening movement are not assumed to be active series.

Appendix B

RQ1 Feature Engineering and Model Specification

B.1 Historical Feature Definitions

For a series with observed daily sales y_t, all features are calculated using observations strictly before the forecast origin t. The principal feature definitions are:

$$\mathrm{Lag}_k(t) = y_{t-k}, \qquad k \in \{1, 7, 14, 28, 56\}.$$

For a rolling window of length w,

$$\mathrm{Mean}_w(t) = \frac{1}{w}\sum_{j=1}^{w} y_{t-j}, \qquad \mathrm{Sum}_w(t) = \sum_{j=1}^{w} y_{t-j}.$$

The rolling standard deviation is calculated over the same historical window. The zero-sales ratio is

$$\mathrm{ZeroRatio}_w(t) = \frac{1}{w}\sum_{j=1}^{w} \mathbb{1}(y_{t-j} = 0).$$

Let N_t^+ denote the number of positive-sales days before origin t, and let T_t denote the length of the observed history. The average demand interval is

$$\mathrm{ADI}_t = \frac{T_t}{N_t^+},$$

when at least one positive-sales day exists. Positive-demand mean and variance are calculated only over positive observations. The squared coefficient of variation is

$$CV_t^2 = \frac{\mathrm{Var}(y_t \mid y_t > 0)}{\mathrm{E}(y_t \mid y_t > 0)^2},$$

when the positive-demand mean is non-zero. Days since last sale is the number of complete days between the most recent positive observation and the forecast origin.

B.2 Context Feature Construction

The context challenger adds historical aggregates without changing the forecast target. For shop b, category c, and window w, the shop aggregate is

$$S_{b,w}(t) = \sum_{j=1}^{w} \sum_{c'} y_{b,c',t-j},$$

and the category aggregate is

$$C_{c,w}(t) = \sum_{j=1}^{w} \sum_{b'} y_{b',c,t-j}.$$

The network aggregate is

$$N_w(t) = \sum_{j=1}^{w} \sum_{b'} \sum_{c'} y_{b',c',t-j}.$$

The model also receives recent shop/category trend summaries and the focal series' share of its shop and category totals. These features describe relative scale and recent movement across related series. All aggregate windows terminate at $t-1$; no observation from the forecast target window is included.

The context experiment is an ablation of information availability: the baseline receives own-series history and calendar features, while the challenger receives those same features plus cross-series context. The estimator family, hyperparameters, target, origins and evaluation rows are held constant.

B.3 Direct Multi-Horizon Target

For each horizon $h \in \{1, 7, 14, 28\}$, the target is the sum of observed units in the next h days:

$$Y_{s,t,h} = \sum_{j=0}^{h-1} y_{s,t+j}.$$

A direct model is fitted for each horizon. This means the one-day target and 28-day target are different prediction tasks. A lower WAPE at a longer cumulative horizon must not be interpreted as equivalent accuracy for daily replenishment.

Historical training origins are sampled on a fixed stride. A training sample is admitted only when its full target window ends before the validation origin. The minimum historical feature window is 56 days. This chronological condition is applied to both the baseline and context-aware models.

B.4 Model Configuration and Reproducibility

The context experiment uses LightGBM Tweedie with variance power 1.3, 100 boosting iterations, learning rate 0.035, 31 leaves, minimum child samples 80, row and column subsampling of 0.85, subsampling frequency 1, L2 regularisation 2.0 and random seed 42. The configuration is fixed between baseline and context model.

The seven-day experiment uses nine monthly origins from January through September 2025. The one-, 14- and 28-day comparisons use July, August and September 2025. Each origin contains all 512 shop–category series. The previously inspected Q4 2025 window is excluded from model selection. The run manifest records the input checksum, evidence status, origins, horizons, feature groups, parameters, bootstrap settings and the fact that no untouched final test has been evaluated.

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